



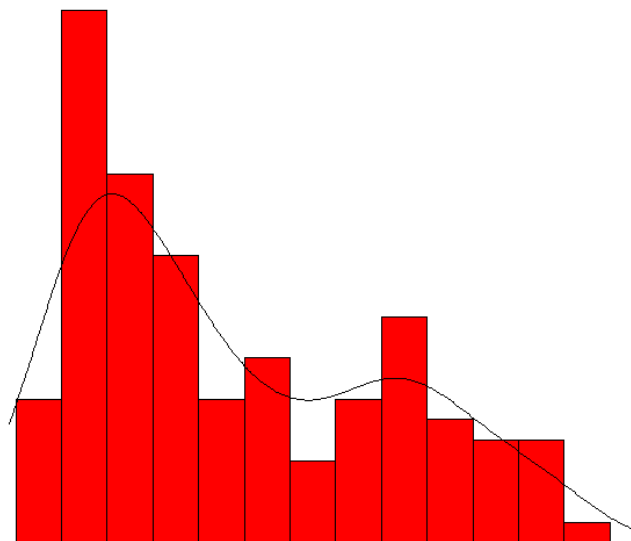
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Deep Learning

INTRODUCTION TO MULTIMODAL LEARNING

Cigdem Beyan
A. Y. 2025/2026

WHAT IS MULTIMODAL?

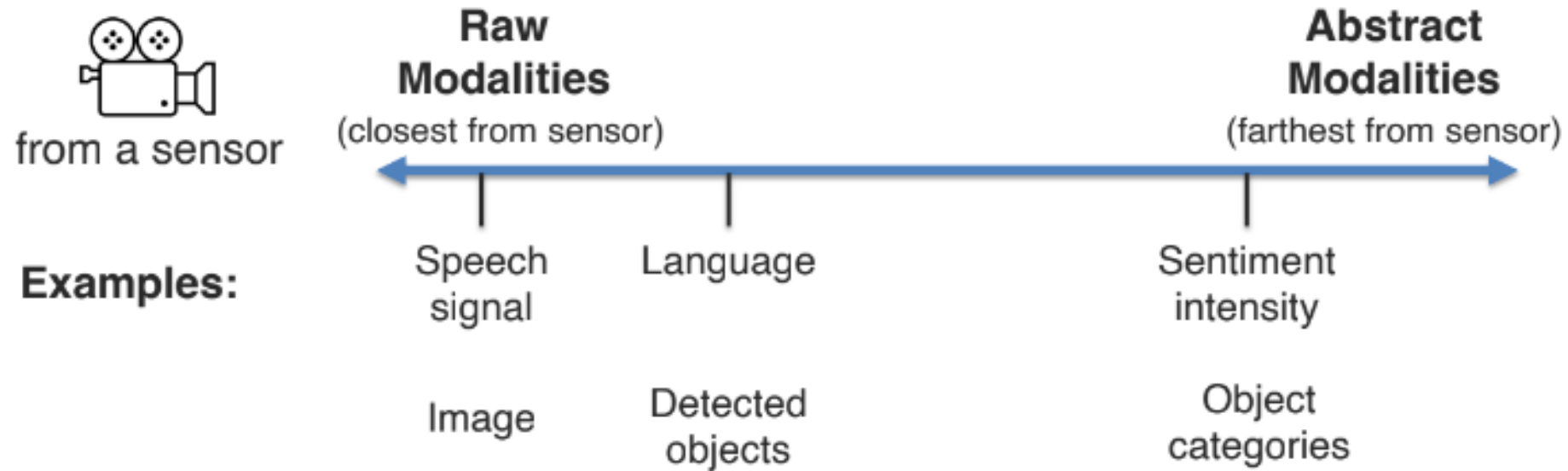


The term multimodality originates from mathematics and refers to having multiple modes or peaks in a probability distribution.

MODALITY



Modality refers to the way in which something expressed or perceived.



MULTIMODALITY

In simpler terms, multimodality refers to the various ways in which we perceive and express information through different sensory modalities such as language, vision, speech, and others.





MULTIMODALITY

A dictionary definition...

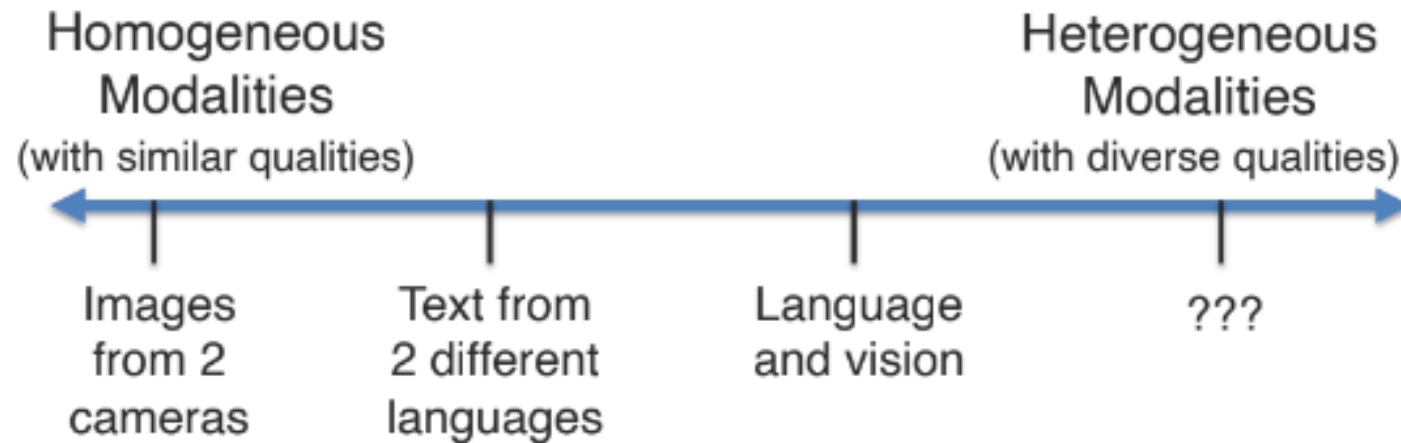
Multimodal: with multiple modalities

A research-oriented definition...

***Multimodal* is the science of
heterogeneous and interconnected data**

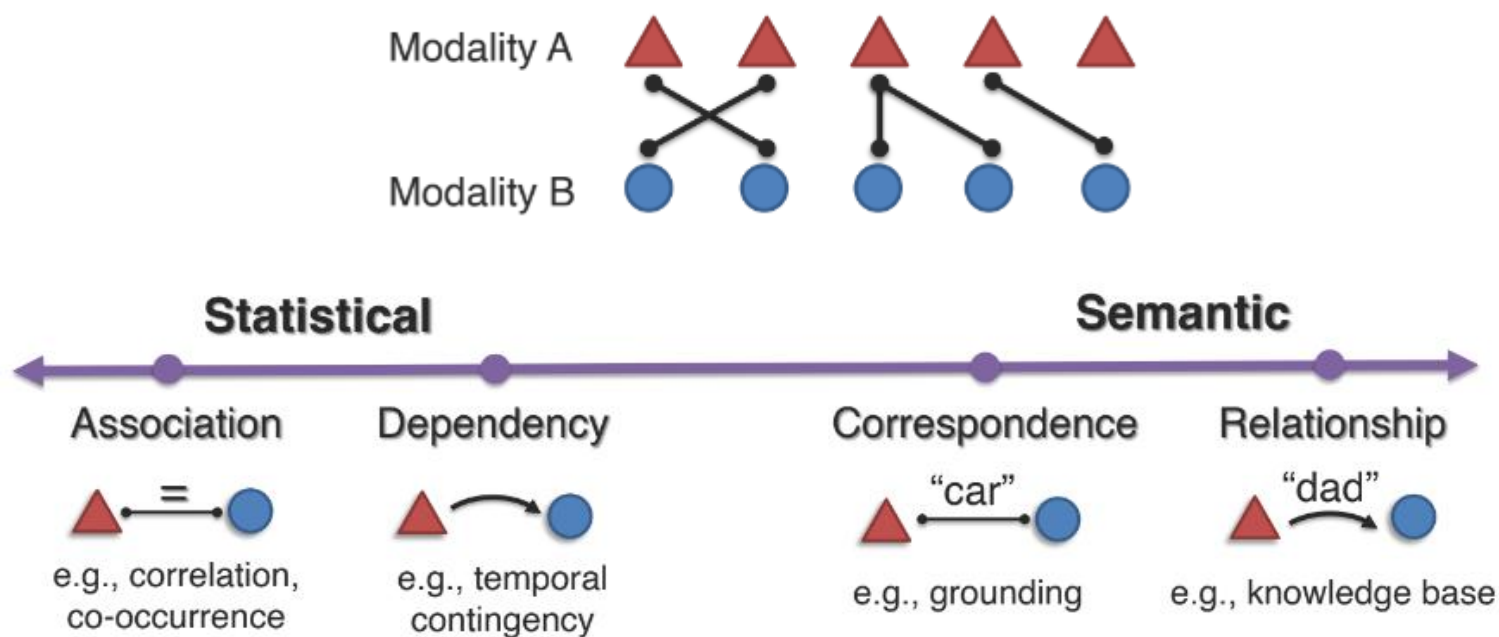
HETEROGENEOUS MODALITIES

- Information present in different modalities will often show diverse qualities, structures and representations.



INTERCONNECTED MODALITIES

- Modalities are often related and share complementary information that interact



typical symptom most common
neuroimaging managing
dysregulation
alleviating
treating complication
distribution
diagnostic disorder
emotional problems atypical mental health
central nervous system
health
difficulty
cognitive
emotions
symptoms occurring factors insufficient
advanced stages sensory psychological diagnosis
depression
genetic origin
mental disorder primary rigidity
doctor pills advanced stages slowness biology environment
different pathological clinical
detailed
spirituality
according
meaning
cause
related
level
definition
disease
illness
tendency
dopamine
drugs
inclusions
clinical
philosophies
ineffective
confirmation
disturbance
well-being
mental
social
expression
movement
related
disorder
tests
mind
brain
degenerative
surgery
effective
pathological
treatment
evidence
movement
symptoms
direction
characterized
clinical
symptoms
acquired
against
disorder
tendency
dopamine
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tests
mind
brain
degenerative
surgery
effective
pathological
treatment
evidence
movement
symptoms
direction
characterized
clinical
symptoms
acquired
against
disorder

[illegible]



(COMMON) MODALITIES

- Natural language – spoken, written
- Visual – images, videos, RGB, infrared (thermal), depth...
- Audio – voice, sound, music
- Biological signals – Electroencephalogram (EEG),
Electrocardiogram (ECG) and Galvanic Skin Response (GSR)
- Haptics – touch
- Motion capture system's data
-

Language

- **Lexicon**
 - Words
- **Syntax**
 - Part-of-speech
 - Dependencies
- **Pragmatics**
 - Discourse acts

Acoustic

- **Prosody**
 - Intonation
 - Voice quality
- **Vocal expressions**
 - Laughter, moans

Visual

- **Gestures**
 - Head gestures
 - Eye gestures
 - Arm gestures
- **Body language**
 - Body posture
 - Proxemics
- **Eye contact**
 - Head gaze
 - Eye gaze
- **Facial expressions**
 - FACS action units
 - Smile, frowning

Touch

- Haptics
- Motion

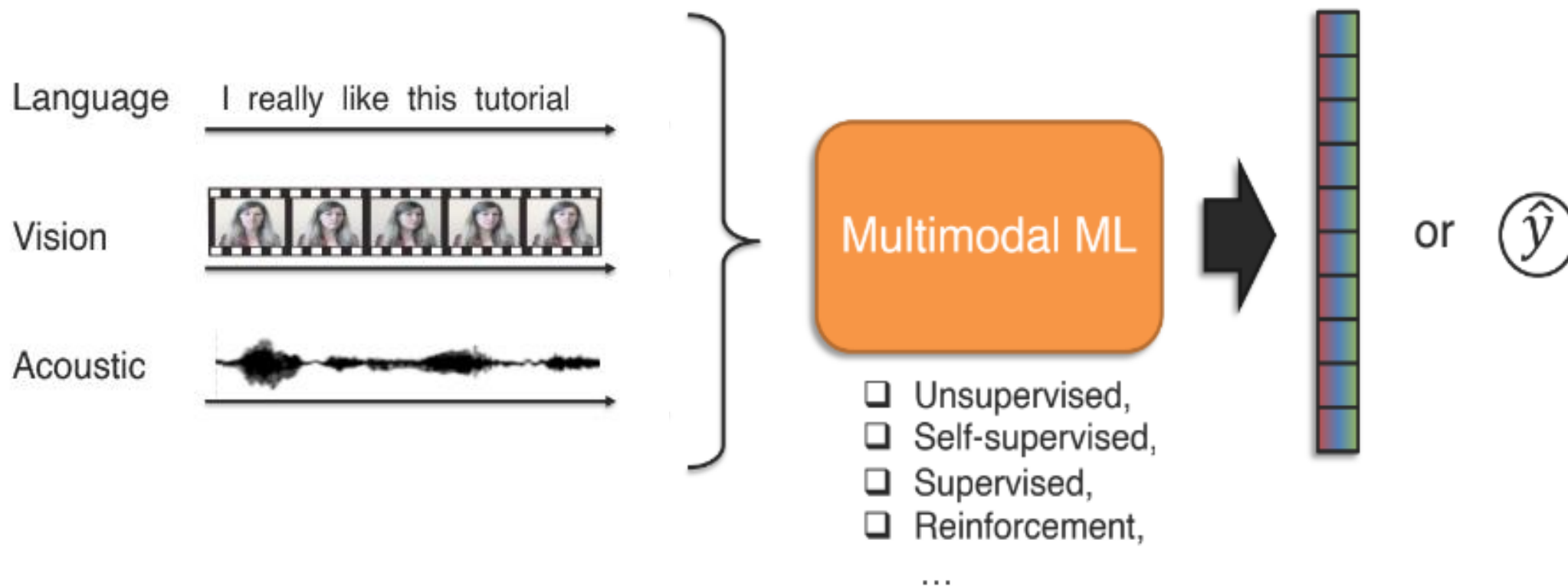
Physiological

- Skin conductance
- Electrocardiogram

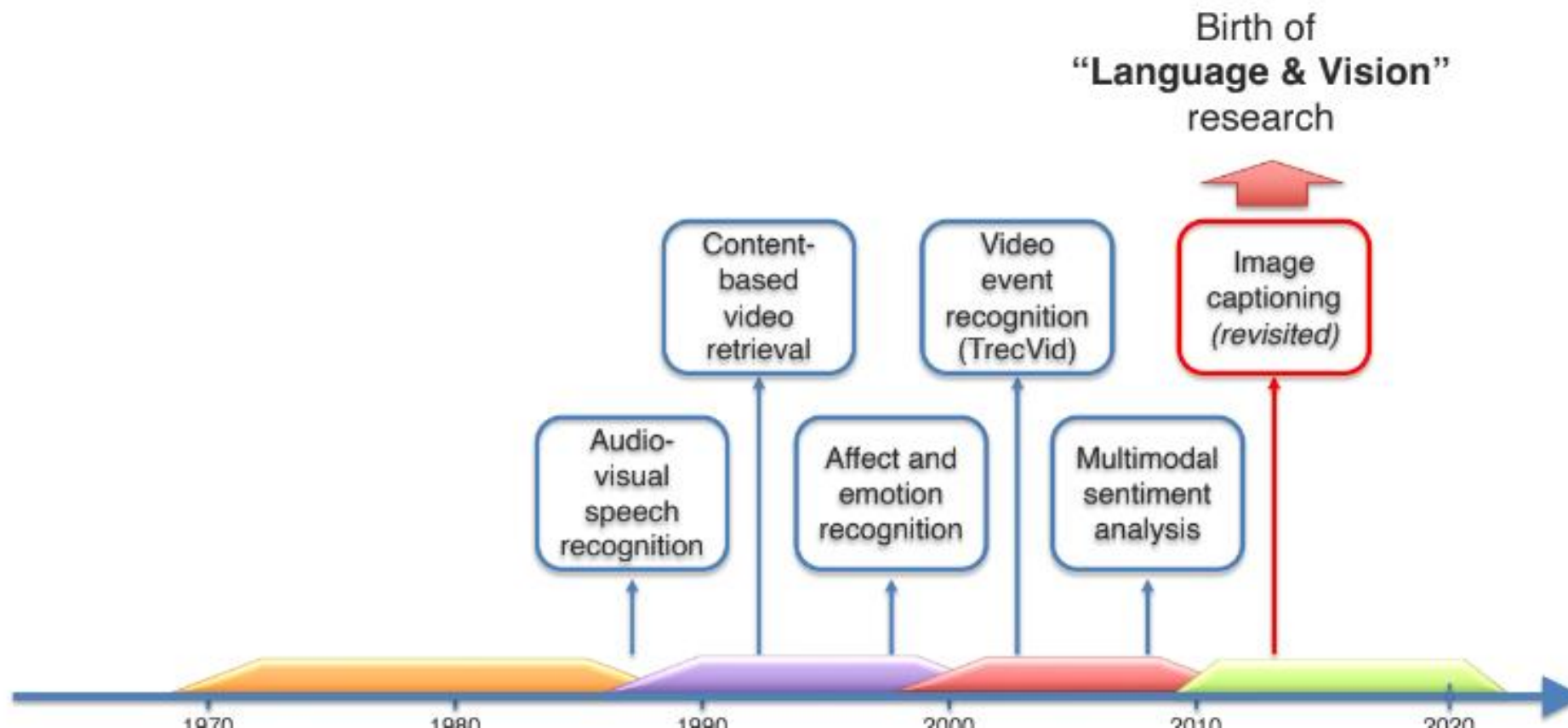
Mobile

- GPS location
- Accelerometer
- Light sensors

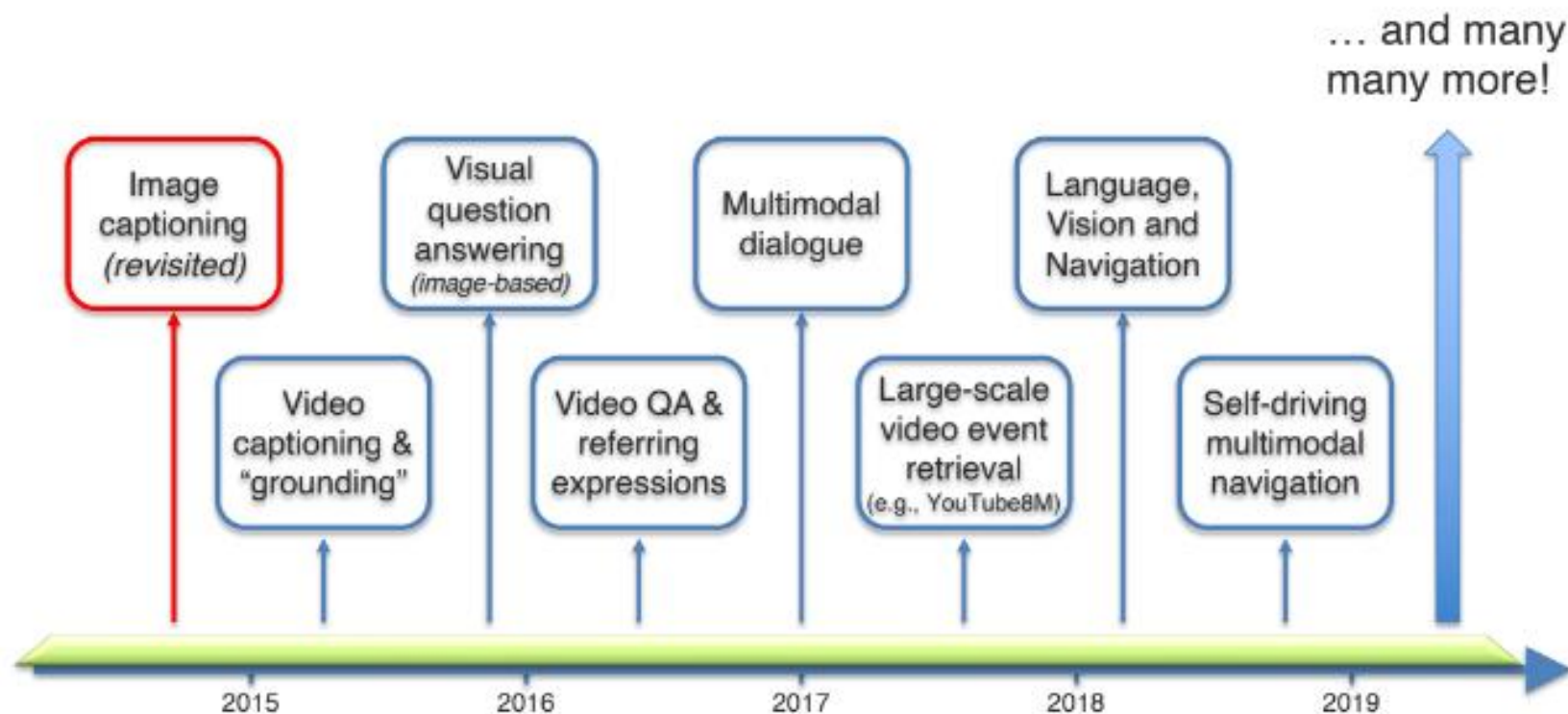
WHAT IS MULTIMODAL MACHINE LEARNING?



AUDIO-TEXT-VIDEO/IMAGE RESEARCH TASKS

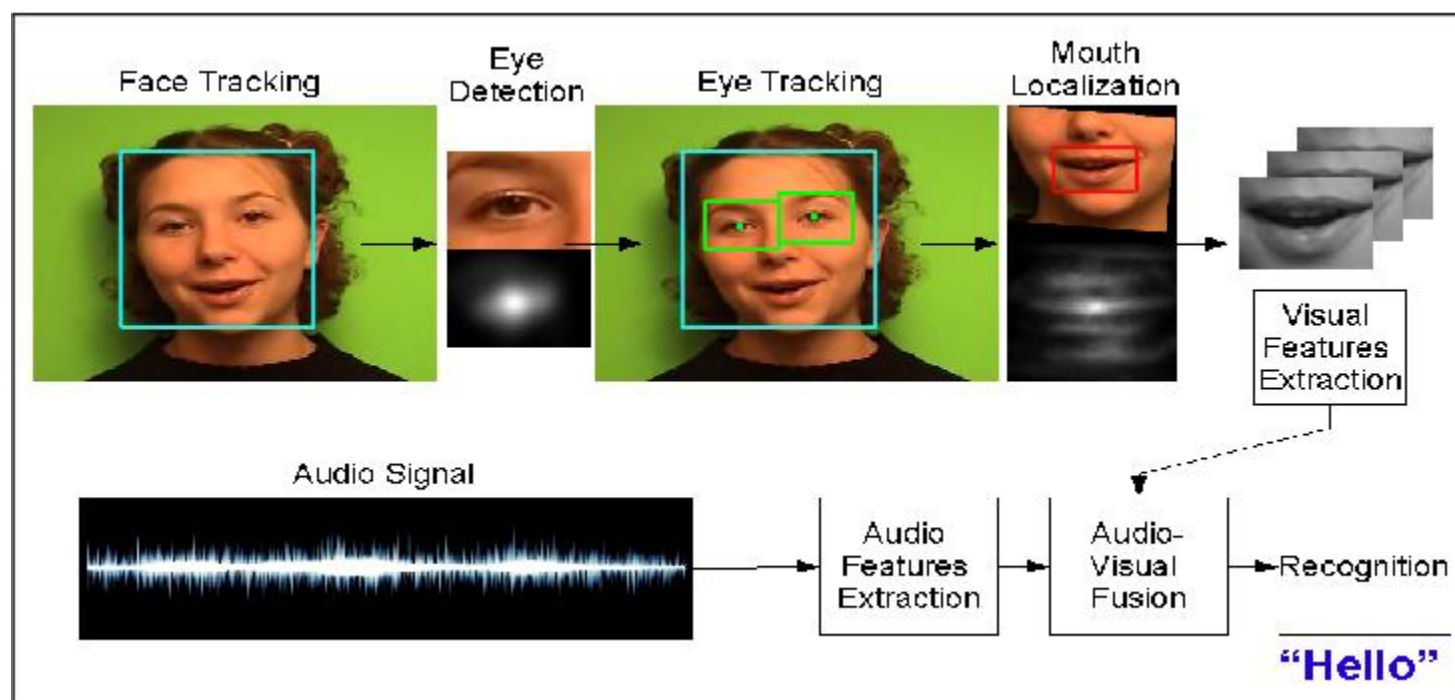


AUDIO-TEXT-VIDEO/IMAGE RESEARCH TASKS



EXAMPLES

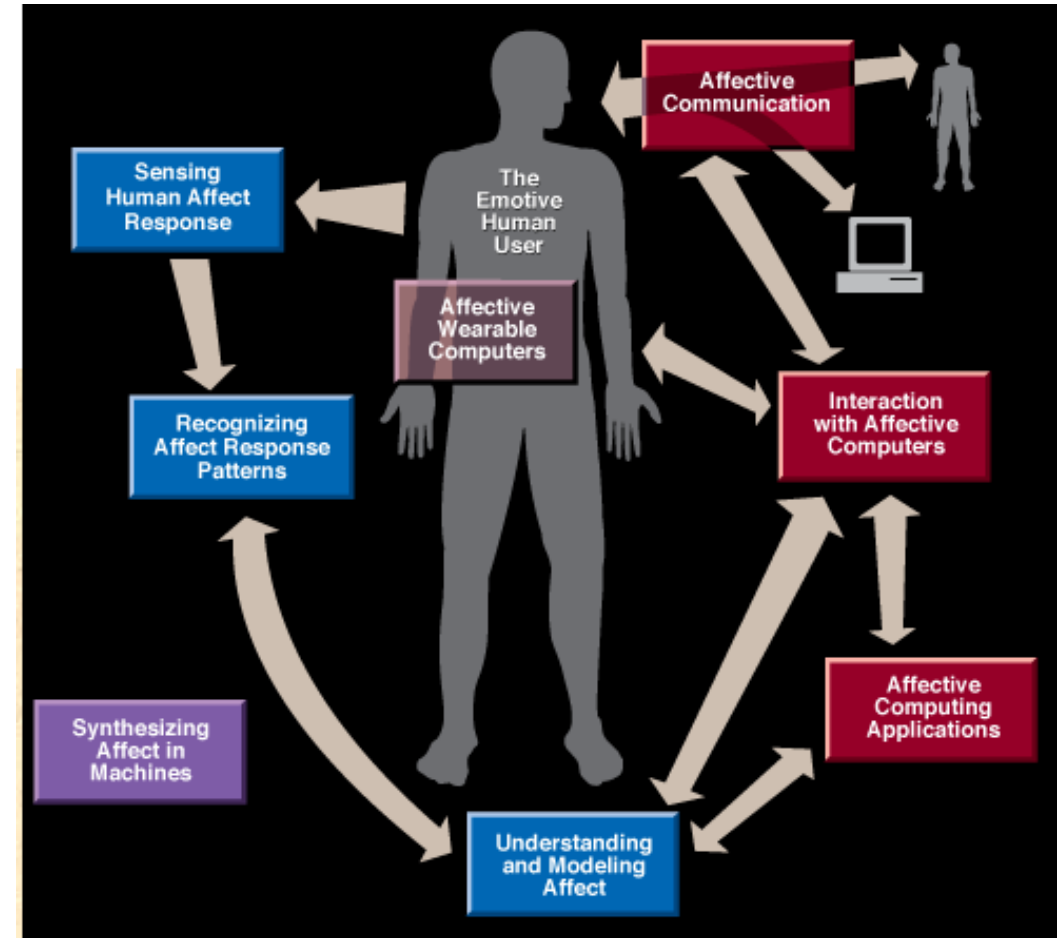
- *Audio-visual speech recognition*



Speech recognizers based on **audio cues** only are sensitive to noise, therefore, **audio-visual** speech recognition systems have become popular.

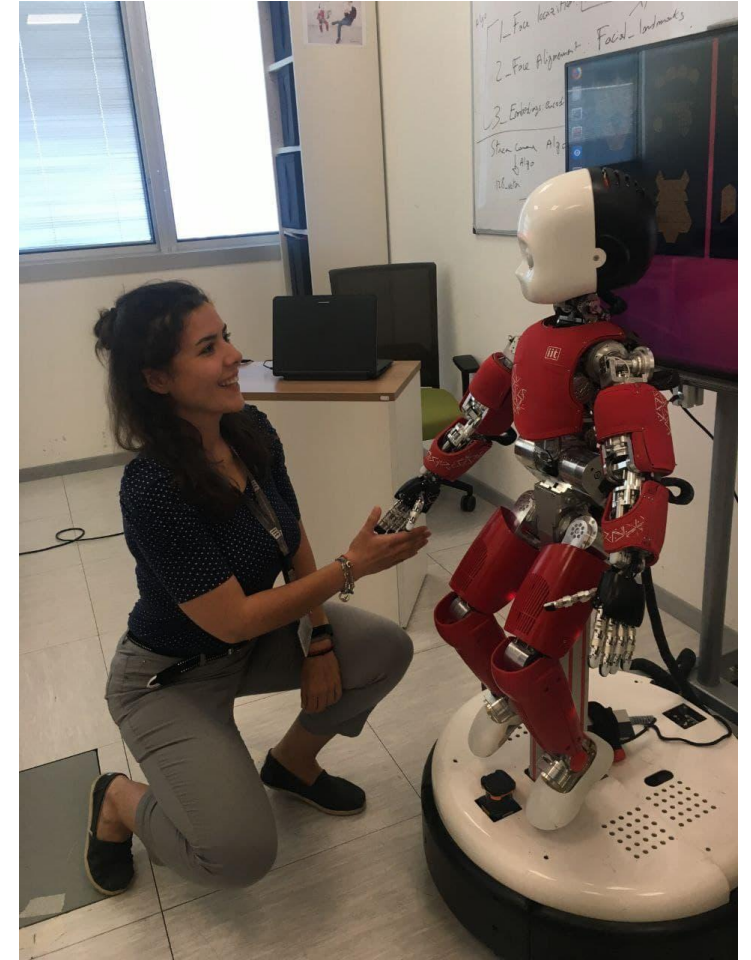
EXAMPLES

- *Affective Computing*
 - Is the study and development of systems and devices that can **recognize**, **interpret**, **process**, and **simulate** human emotions and other internal states.
 - Why important? Emotions are key for decision making!

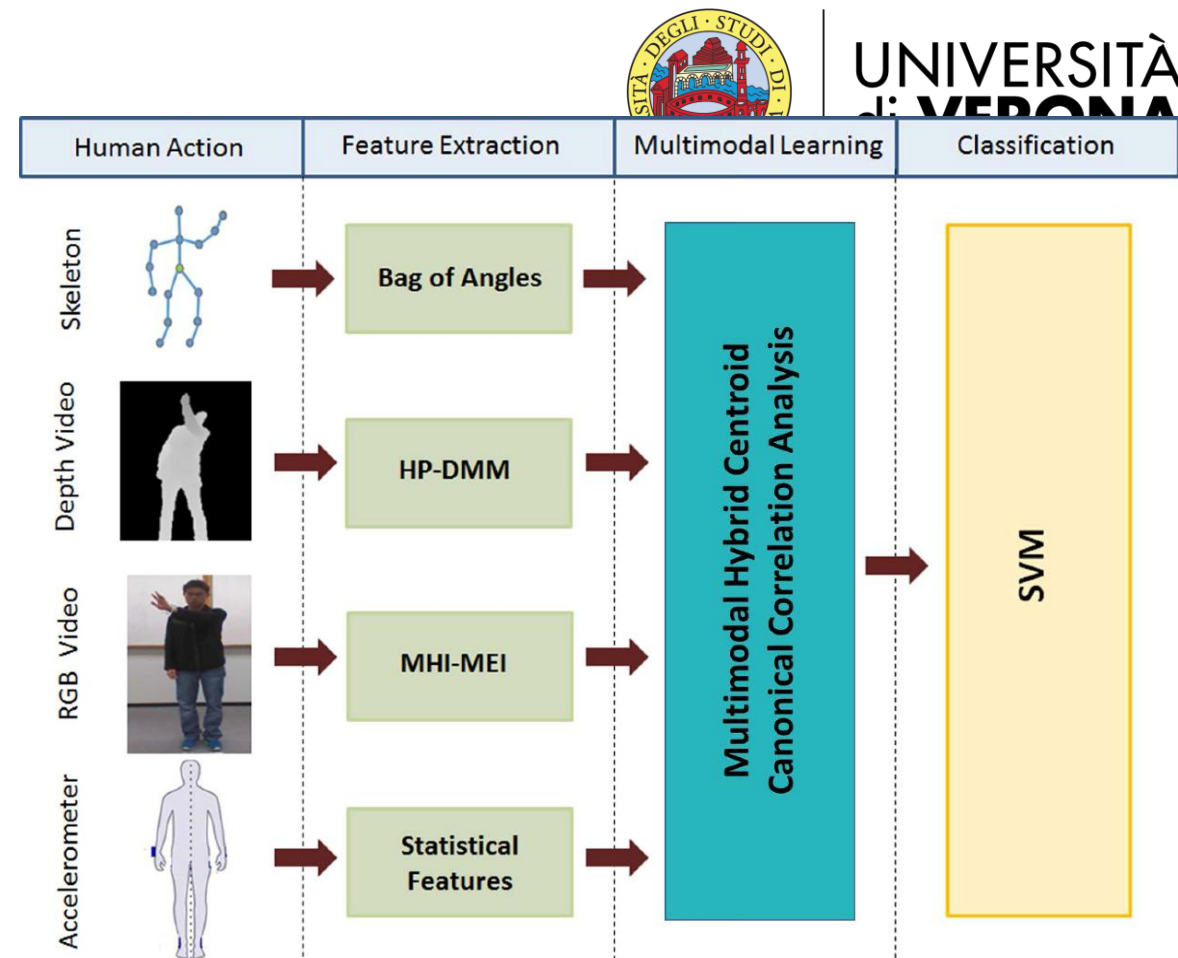
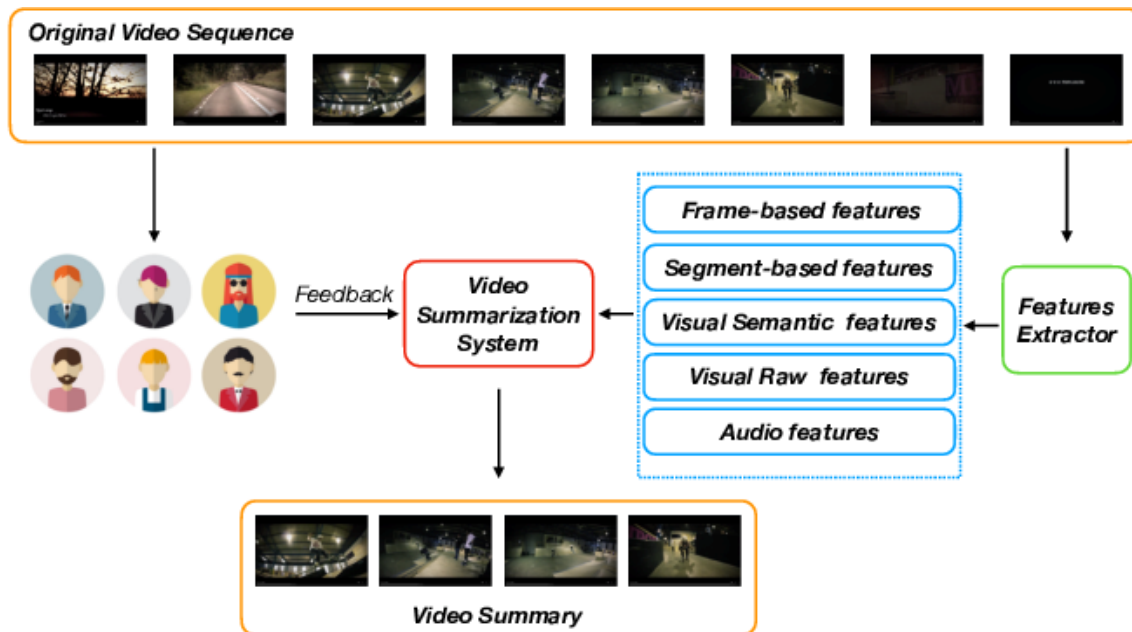


EXAMPLES

- *Human- human- robot interaction analysis*
 - Social Signal Processing (social AI)
 - Integrates social psychology concepts into Artificial Intelligence and focuses on **automatic detection**, **interpretation**, and **synthesis** of **social signals** represented in terms of **nonverbal behavioral cues** capable of conveying socially-relevant information.



EXAMPLES



- Event / Behavior Understanding
 - Action recognition
 - Event segmentation
 - Video summarization

EXAMPLES

- *Content Understanding*
 - Is the ability of computers to **ingest**, **relate**, and **process information contained in data** and to **perform tasks** using that information.
 - Considers all types of data, e.g., text, speech, images, sensor data, and static or streaming data.

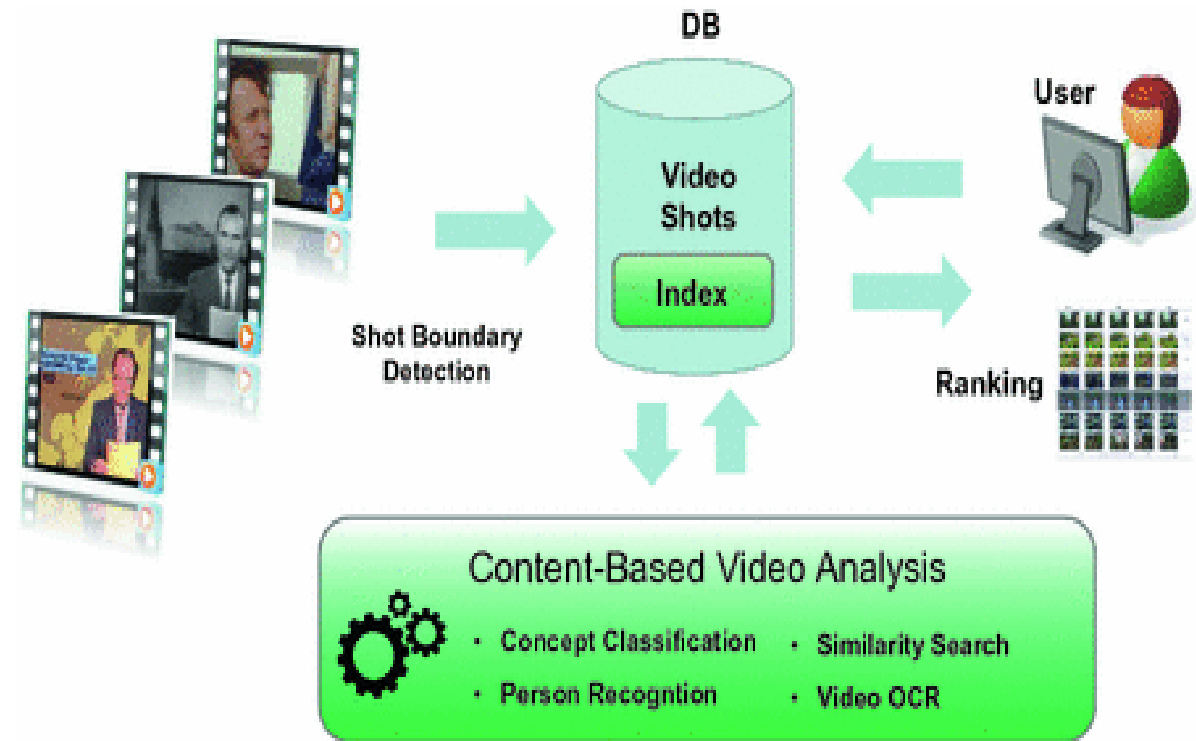


Image credit: Mühling et al., Content-Based Video Retrieval in Historical Collections of the German Broadcasting Archive, 2016.

EXAMPLES



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- *Multimedia information retrieval*
 - Content understanding
 - Image captioning
 - Video captioning
 - Visual Question Answering



a little girl sitting on a bench holding an umbrella.



a herd of sheep grazing on a lush green hillside.



a close up of a fire hydrant on a sidewalk.



a yellow plate topped with meat and broccoli.



a zebra standing next to a zebra in a dirt field.



a stainless steel oven in a kitchen with wood cabinets.



two birds sitting on top of a tree branch.

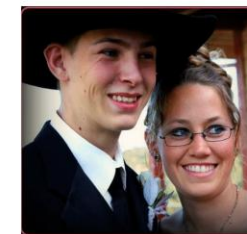


an elephant standing next to rock wall.

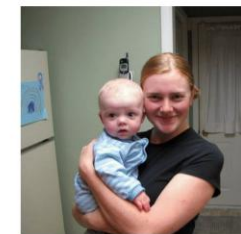


a man riding a bike down a road next to a body of water.

Who is wearing glasses?
man woman



Where is the child sitting?
fridge arms



Is the umbrella upside down?
yes no



How many children are in the bed?
2 1





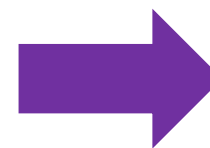
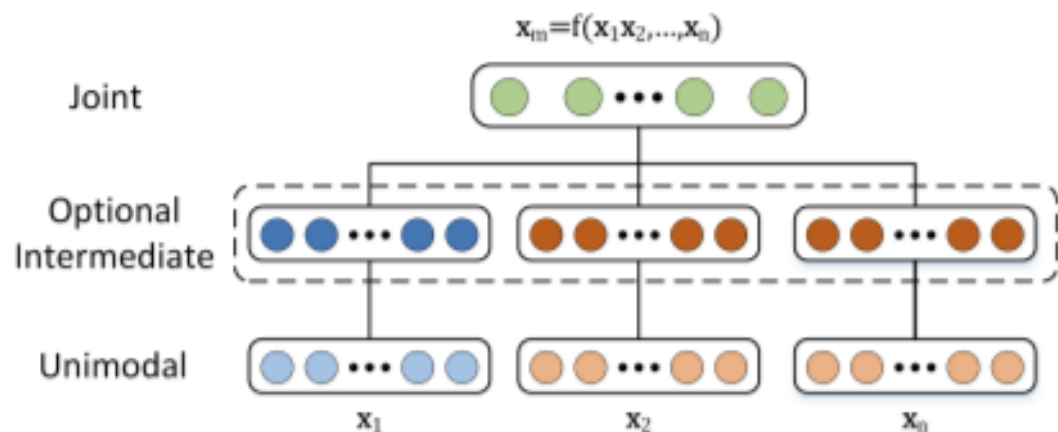
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MULTIMODAL TECHNICAL CHALLENGES & TASKS

REPRESENTATION

- Learning representations that reflect cross-modal interactions between individual elements, across different modalities.
- How to represent and summarize multimodal data in a way that exploits the **complementarity** and **redundancy** of multiple modalities.

Joint representations

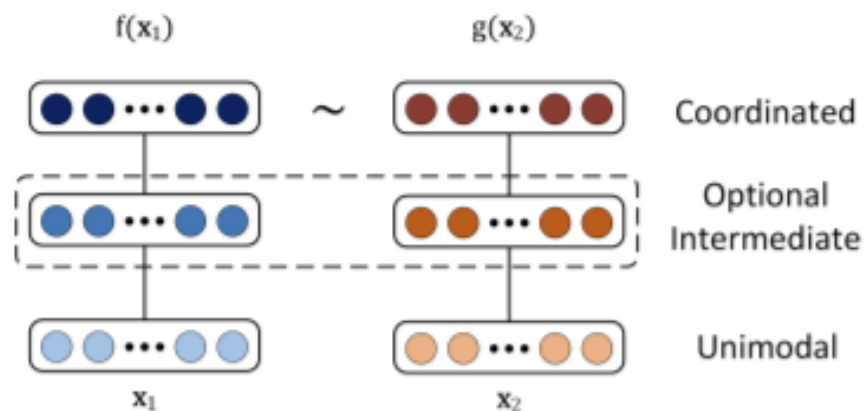


Joint representations are projected to the same space using all of the modalities as the input.

REPRESENTATION

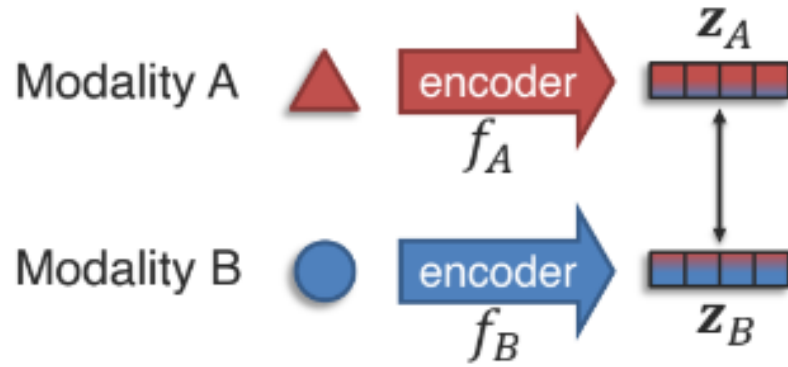
Coordinated representations

- Projecting all the modalities to their space, but those spaces are coordinated using a constraint.
- This kind of approach is more useful for modalities that are fundamentally very different and might not work well in a joint space.



Unimodal representations exist in their own space, but are coordinated through a similarity (e.g. cosine distance).

COORDINATION W/ CONTRASTIVE LEARNING



Contrastive loss:

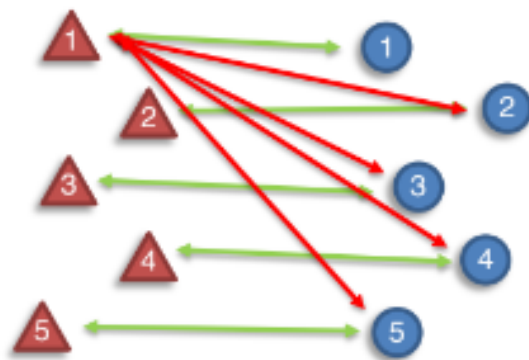
→ brings **positive pairs** closer and pushes **negative pairs** apart

Simple contrastive loss:

$$\max\{0, \alpha + \underbrace{\text{sim}(z_A, z_B^+)}_{\text{positive pairs}} - \underbrace{\text{sim}(z_A, z_B^-)}_{\text{negative pair}}\}$$

Similarity functions are often cosine similarity

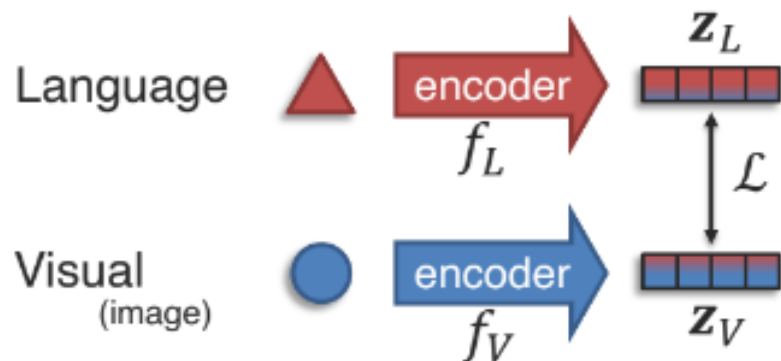
Paired data: $\{\triangle, \circ\}$
(e.g., images and text descriptions)



←→ Positive pairs
←→ Negative pairs

CLIP (CONTRASTIVE LANGUAGE-IMAGE PRE-TRAINING)

**noise
contrastive
estimation
loss**



Popular contrastive loss: InfoNCE

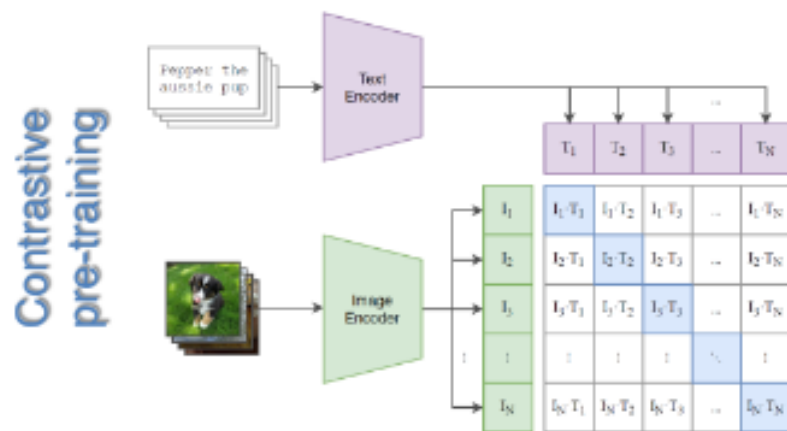
$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^i)}{\sum_{j=1}^N \text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^j)}$$

Similarity function can be cosine similarity

positive pairs

negative pairs and positive pairs

Positive and negative pairs:



→ f_L and f_V are great encoders for language-vision tasks

→ $\mathbf{z}_L$ and $\mathbf{z}_V$ are coordinated but not identical representation spaces



CLIP (CONTRASTIVE LANGUAGE-IMAGE PRE-TRAINING)

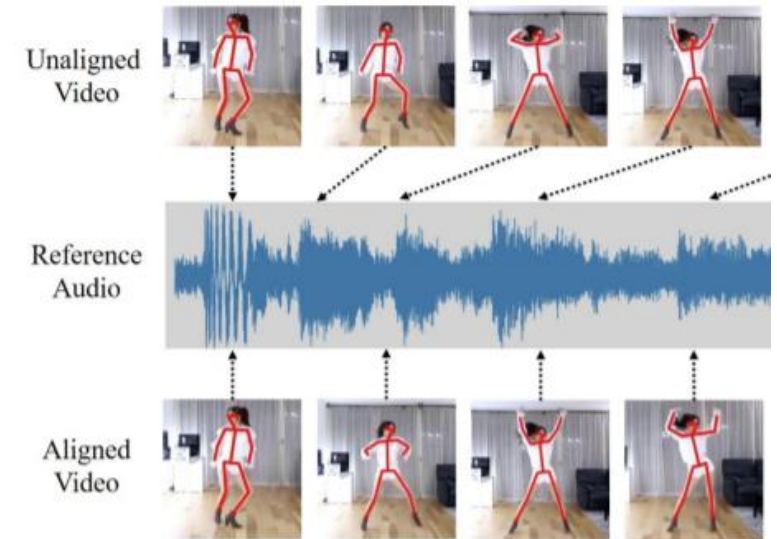
- CLIP is still very popular, and the main reason is its **zero-shot** capability and **generalizability**.
- It has been applied to several fields. As an “unexpected field” **medical imaging** can be given.
 - Classification, zero-shot classification, segmentation, detection, image-text retrieval, visual question answering, report generation,
 - X-ray, ECG, Histology, CT, Dermatology, MRI, Ultrasound....
- We will come back to CLIP to investigate it in detail....!

ALIGNMENT

- Finding **relationships** and **correspondences** between sub-components of instances from two or more modalities. Identifying connections between elements of multiple modalities.
- Ex:
 - Given an image and a caption, we want to find the areas of the image corresponding to the caption's words or phrases.
 - Given a movie, align it to the script or the book chapters.

EXPLICIT ALIGNMENT

- We are explicitly interested in aligning sub-components between modalities. Alignment is the task in itself.
- Finding correspondence between elements of different modalities, typically through a **loss function**.
- Finding similar data across modalities. E.g., aligning speech signal to a transcript.
- Methods: e.g., **dynamic time warping**



IMPLICIT ALIGNMENT

- Used as an intermediate (often latent) step for another task.
- This allows for better performance.
- Method: *e.g.*, **attention mechanism** (as well as **self-attention** and **transformers**), which allows the decoder to focus on sub-components of the source instance.
- Example task: image and video captioning, visual question answering, cross-modal retrieval.



KEY BENEFITS OF ATTENTION MECHANISM IN MULTIMODAL DATA

- Enables a model to assign **different weights** to different parts of the input data.
 - The model can focus more on the relevant parts of each modality while ignoring less important information.
 - Ex: in a task involving both text and images, the model can learn to focus on the parts of the image that correspond to certain words or phrases in the text.



KEY BENEFITS OF ATTENTION MECHANISM IN MULTIMODAL DATA

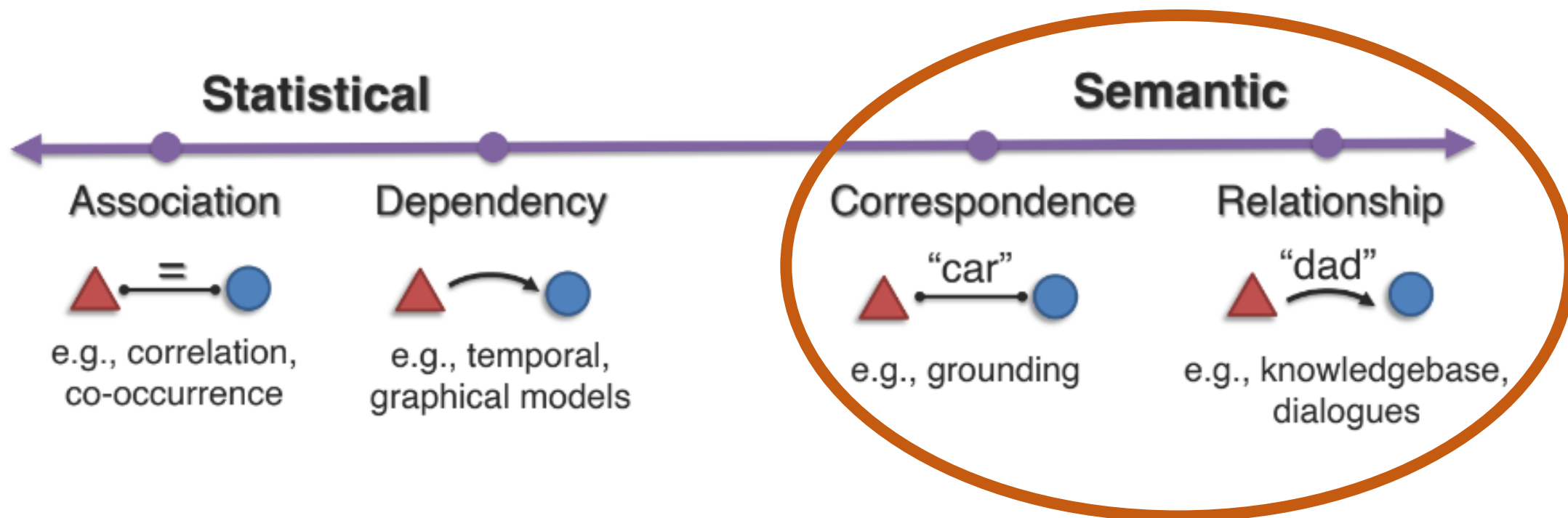
- Effective in dealing with sequences of **varying lengths**.
 - Important for multimodal data where, for example, the text description length may differ from the duration of the corresponding audio or the number of objects in an image.



KEY BENEFITS OF ATTENTION MECHANISM IN MULTIMODAL DATA

- Different modalities may have different structures and temporal alignments. Attention mechanisms help align these modalities by learning **which parts of one modality correspond to which parts of another modality**.
 - This is essential for tasks like image captioning (aligning image regions with words) and video description (aligning video frames with text).
- Recall that SOTA models like **BERT** for text, **ViT** for images, and multimodal models like **CLIP** all use attention mechanisms to integrate and align info from different sources effectively.

ALIGNMENT



GROUNDING

- Establishing **correspondences** between different modalities. These tasks are fundamental for creating models that understand and integrate information from multiple sources, **making the information in one modality meaningful in the context of another.**



Visual grounding refers to the task of linking specific regions or objects in an image or video with corresponding textual descriptions or linguistic elements. It involves identifying and localizing parts of the visual content that correspond to words or phrases in the text.

GROUNDING

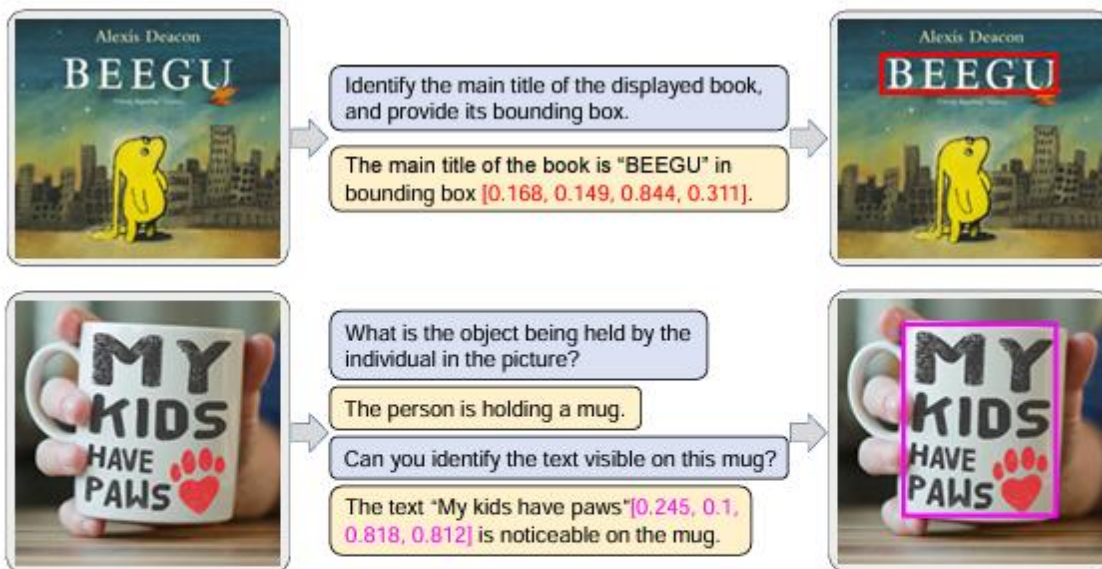


Figure 1. VQA in text-rich images with text-grounding. Given an image and a query, the model generates responses in an autoregressive way, and concurrently outputs the basis of reasoning, *i.e.*, the bounding box that delineates the location of the answer.

Text grounding involves linking words or phrases in a text to corresponding elements in other modalities, such as images, audio, or video.

It is about understanding the context and meaning of text by associating it with relevant non-textual data.

GROUNDING → SEGMENTATION

Referring Expression Segmentation (RES) aims to generate a segmentation mask for the object described by a given language expression.

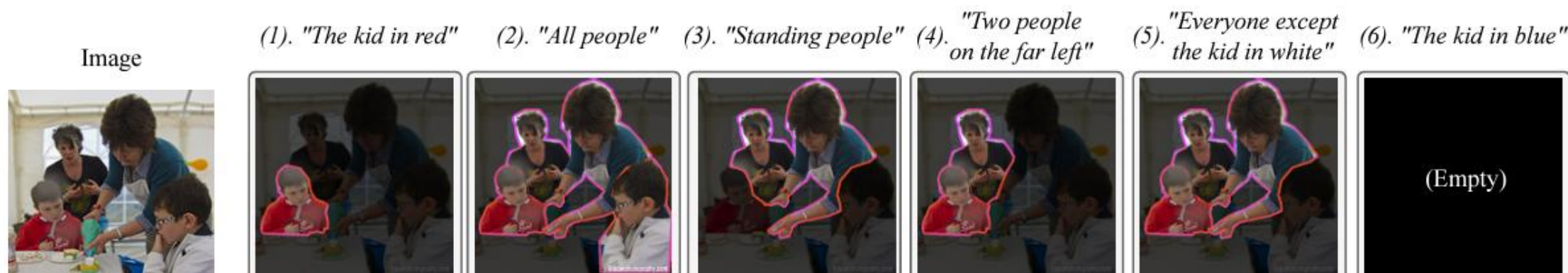


Image (b)

i. "The bike **and** two passengers on it"

ii. "The bike **that has** two passengers and **its** driver"

TRANSLATION

- How to map data from **one modality** to **another modality**.
- Given an entity in one modality the task is to **generate** the same entity in a different modality.
- Ex: given an image we might want to generate a sentence describing it or given a textual description generate an image matching it.



The man at bat readies to swing at the pitch while the umpire looks on.



A large bus sitting next to a very tall building.



A horse carrying a large load of hay and two people sitting on it.



Bunk bed with a narrow shelf sitting underneath it.

Example images and captions from the Microsoft COCO Caption dataset.

TRANSLATION

Example-based models use a *dictionary* when translating between the modalities.

E.g., retrieval-based models (*e.g.*, finding the closest sample in the dictionary)

- Image captioning
- Media retrieval
- ...

TRANSLATION

Example-based models use a *dictionary* when translating between the modalities.

E.g., retrieval-based models (*e.g.*, finding the closest sample in the dictionary)

- Image captioning
- Media retrieval
- ...

Generative models construct a model that can produce a translation.

E.g., Encoder-decoder models: to first encode a source modality into a vectorial representation and then use a decoder module to generate the target modality.

- Text to image/image to text
- Sound synthesis
- Video description generation

TRANSLATION - EXAMPLE

Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks

Jun-Yan Zhu* Taesung Park* Phillip Isola Alexei A. Efros
Berkeley AI Research (BAIR) laboratory, UC Berkeley

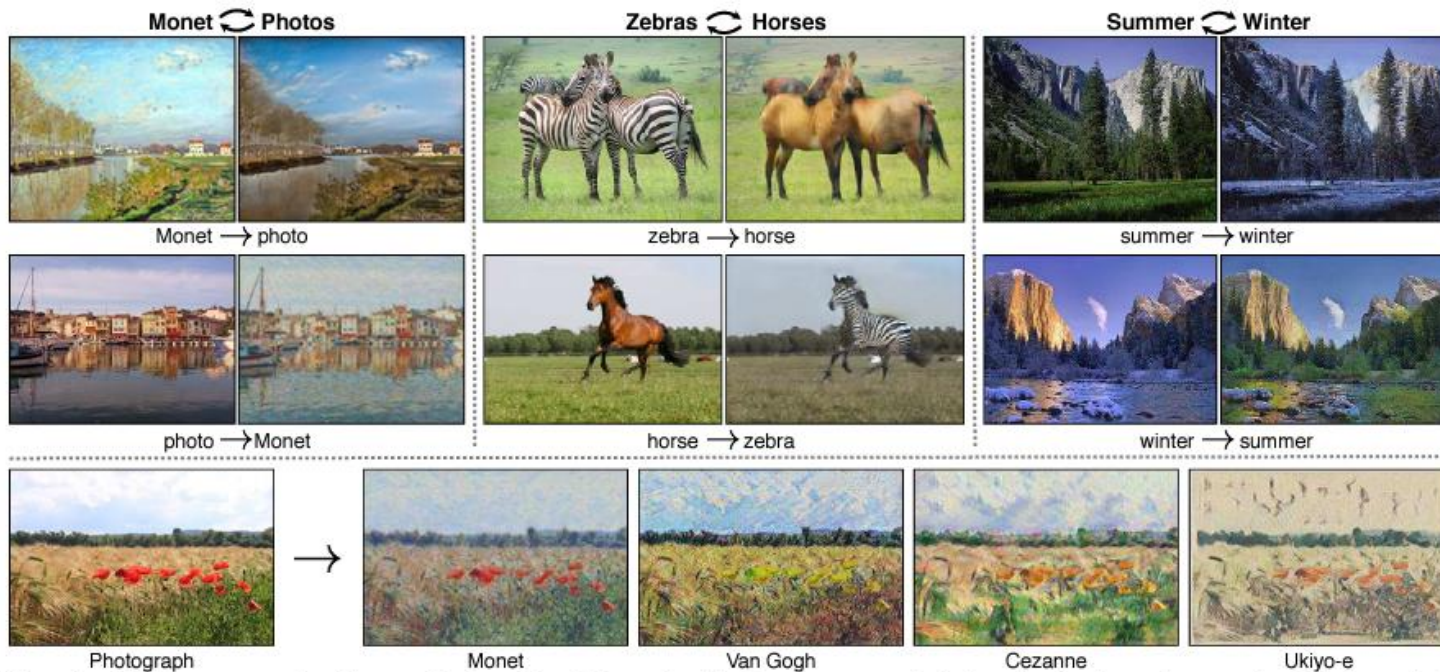


Figure 1: Given any two unordered image collections X and Y , our algorithm learns to automatically “translate” an image from one into the other and vice versa. Example application (*bottom*): using a collection of paintings of a famous artist, learn to render a user’s photograph into their style.

- GANs’ success is the idea of an **adversarial loss** that forces the generated images to be, in principle, indistinguishable from real images.
- This is particularly powerful for image generation tasks, as this is exactly the objective that much of computer graphics aims to optimize.
- They use an adversarial loss to learn the mapping such that the translated image cannot be distinguished from images in the target domain.

TRANSLATION – EXAMPLE 2



(a) Input



(b) Ours



(c) GAN-Adapt.



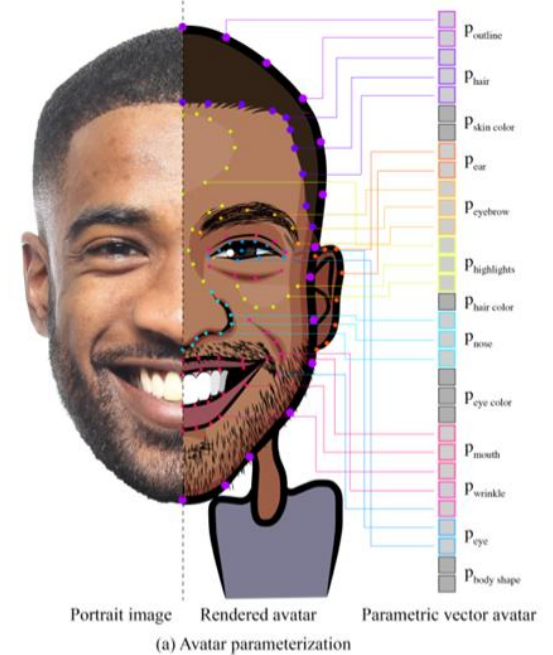
(d) StyleCariGAN



(e) U-GAT-IT

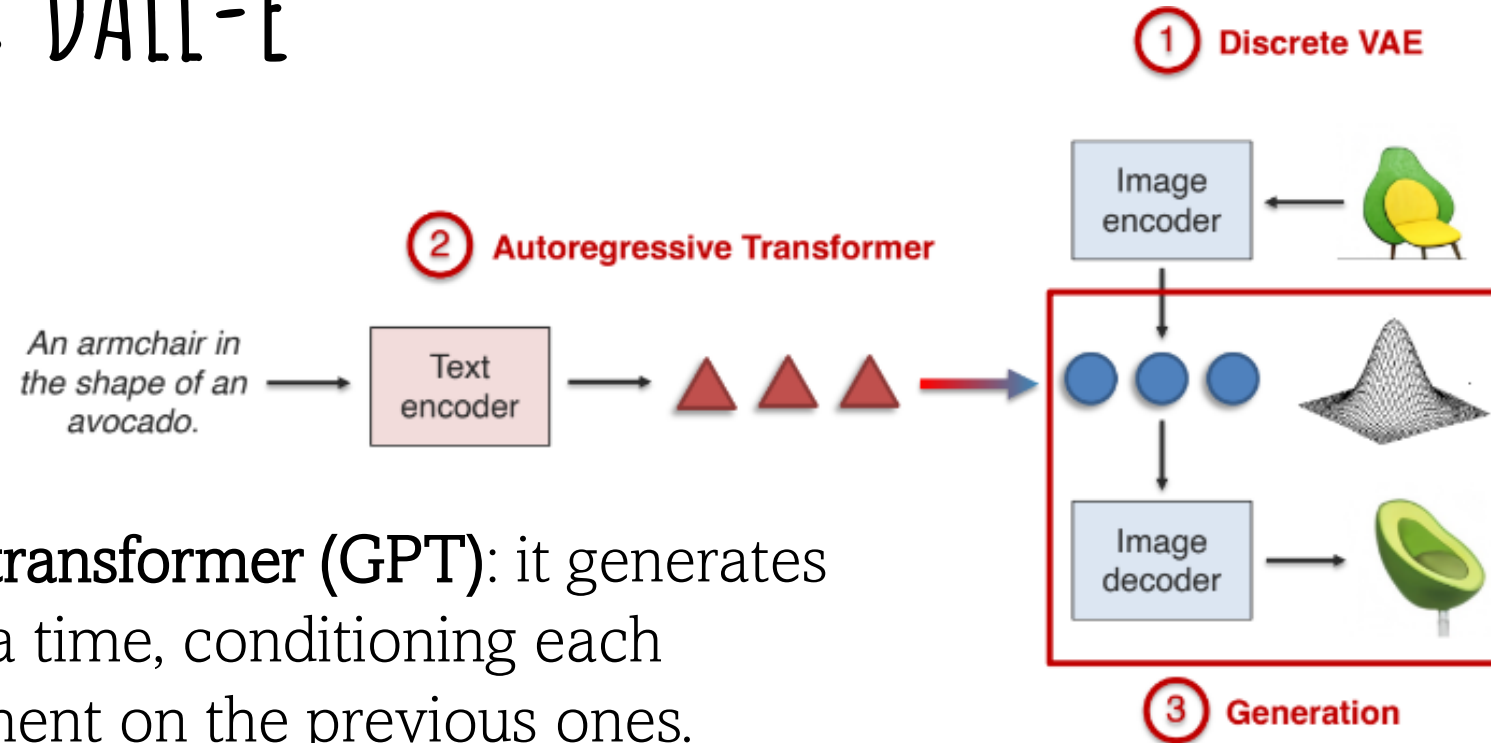


(f) StarGANv2



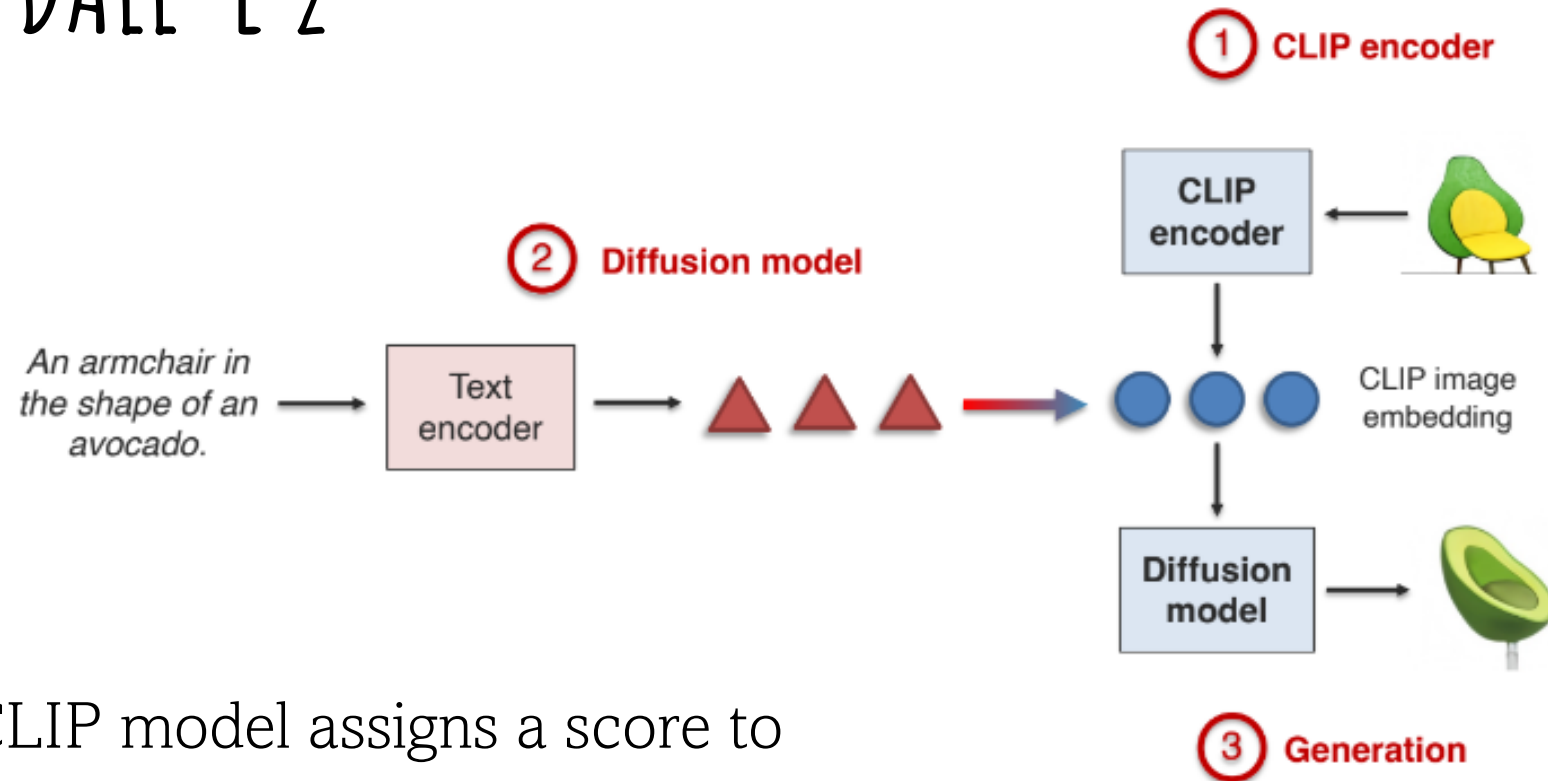
The authors employ a set of parameters that define each facial attribute.

EXAMPLE 3: DALL-E



- **Autoregressive transformer (GPT)**: it generates one element at a time, conditioning each subsequent element on the previous ones.
- The encoder of **dVAE** is responsible for mapping the input data to **discrete codes** in the latent space.
- The decoder network in the **dVAE** is responsible for reconstructing the original data from the generated **discrete codes**.

EXAMPLE 3: DALL-E 2



- In Dall-E , the CLIP model assigns a score to each generated image based on how well it aligns with the textual prompt.
- Instead, Dall-E 2 uses a **diffusion model** conditioned on **CLIP image embeddings**, which, during inference, are generated from **CLIP text embedding** by a prior model.

EXAMPLE 4: STABLE DIFFUSION

- Text-to-Image translation
- Image-to-text captioning (not so preferable, there are better models)
- Image-to-image translation with Text Guidance
- Text-Guided Image Editing
 - Given an image of a forest and the instruction "add a waterfall," the model generates the same forest scene but with a waterfall added.



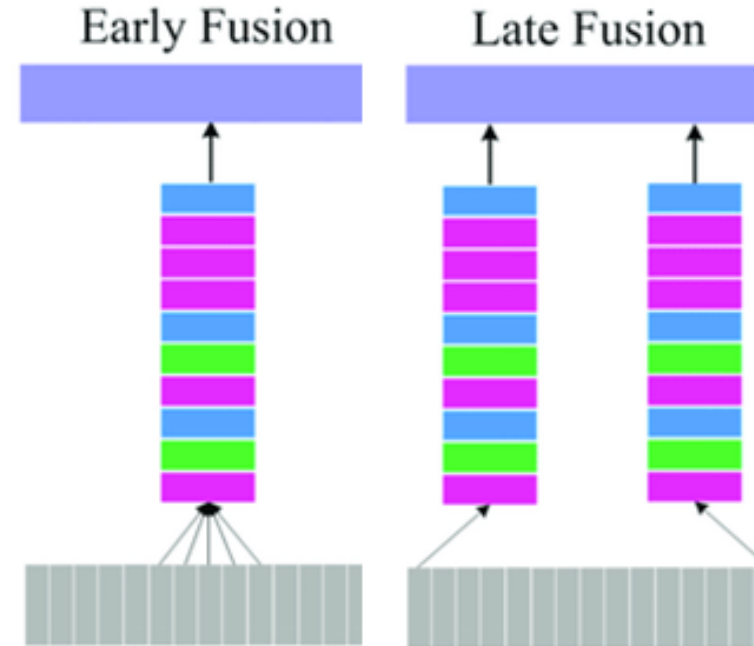
EXAMPLE 4: STABLE DIFFUSION (RECAP)

- Components:
 - **Forward Diffusion Process:** The forward process gradually adds Gaussian noise to an image over several time steps. A clean image is converted to noise.
 - **Reverse Diffusion Process:** to recover the original image by removing the noise step-by-step. This is achieved through a neural network (**U-Net**) trained to predict and subtract noise from the noisy images at each step.
 - **(CLIP) Transformer-based text encoder:** Input textual description is encoded with the encoder and used to condition the U-Net, guiding the image generation process to ensure that the resulting image matches the description.
 - **Latent Space:** Instead of working on image space, it works on lower dimensional latent space.
 - This involves using an encoder (e.g., VAE) to map images to a latent representation, where the diffusion process is applied. After the diffusion process, a decoder reconstructs the image from the latent representation.

FUSION

- How to join two or more modalities to perform a prediction.

- Learn to exploit the correlation and interactions between low level features of each modality.
- Only requires the training of a single model, making the training pipeline easier.



- Uses unimodal decision values and fuses them using e.g., by averaging with voting schemes, weighting based.
- Use of different models for each modality as different predictors can model each individual modality better, allowing for more flexibility.

FUSION

- **Multiple Kernel Learning:** an extension to kernel support vector machines that allow for the use of different kernels for different modalities/views of the data. Kernels can be seen as similarity functions between data points, allowing for better fusion of heterogeneous data.
- **Graphical models:** can be classified as generative: modeling joint probability; or discriminative: modeling conditional probability. Coupled and factorial hidden Markov models, Dynamic Bayesian networks, multi-stream HMM, conditional random fields....

FUSION

- **Neural Networks:** For fusing temporal multimodal information through the use of RNNs and LSTMs, image captioning where a CNN image representation is decoded using an LSTM language model....
- [+] The capacity to learn from large amounts of data.
- [+] Allowing end-to-end training of both the multimodal representation component and the fusion component.
- [+] Better performance when compared to non-NN approaches.
- [-] **Lack of interpretability. Hard to say which modality/feature plays an important role in prediction.**
- [-] Need for large training datasets.

CO-LEARNING

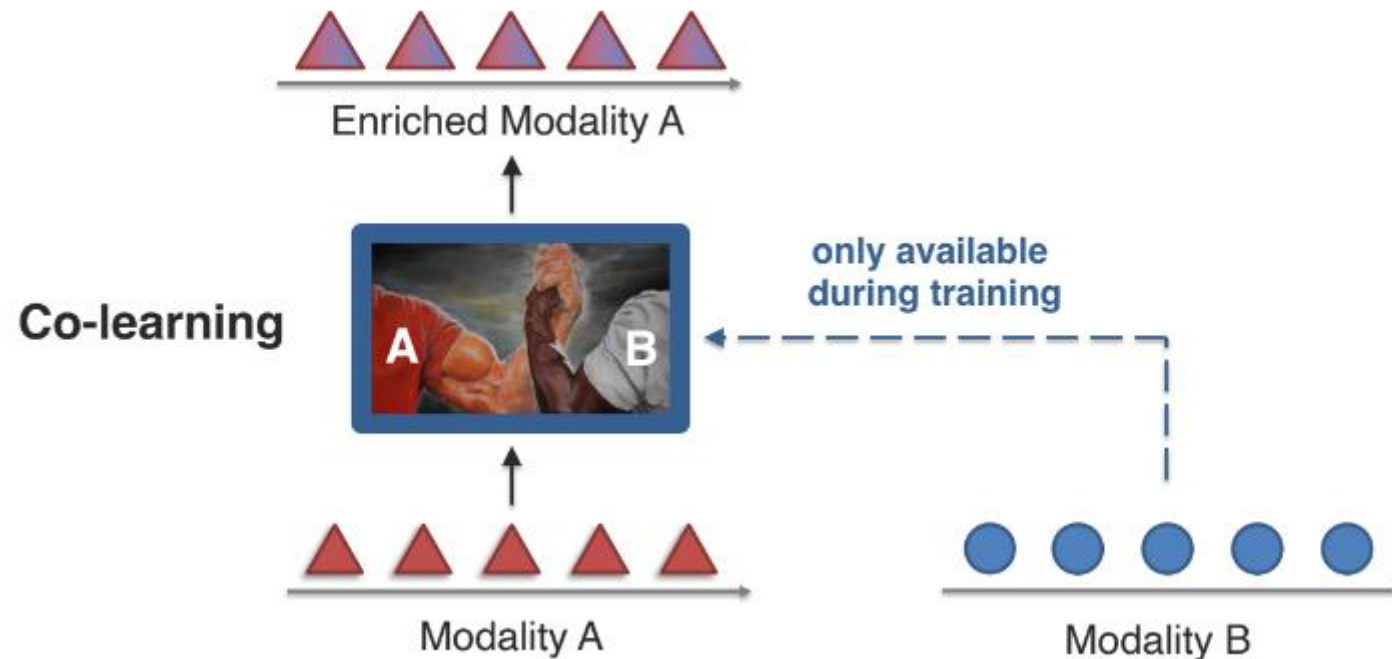
- How knowledge learning from one modality can **help** a computational model trained on a different modality.
- **Transferring knowledge** between modalities, including their representations and predictive models.
- It is particularly relevant when one of the modalities has limited resources — lack of annotated data, noisy input, and unreliable labels.
- As most often the helper modality is used only during model training and is not used during test time.

WHY CO-LEARNING?

Objective	Purpose
Presence of Modality	Should predict when one or more modality is missing fully or partly at test and training time
Data Parallelism	Should support strongly paired, weakly paired and paired through shared modality
Noisy Modality	Should work in noisy conditions in data and labels
Modality Annotations	Should handle annotated, partially annotated and non-annotated modalities during training
Domain Adaptation	Should perform when there is a different dataset/ domain/ modality at training and testing
Interpretability & Fairness	Should provide interpretable and unbiased predictions with explanations and fairness

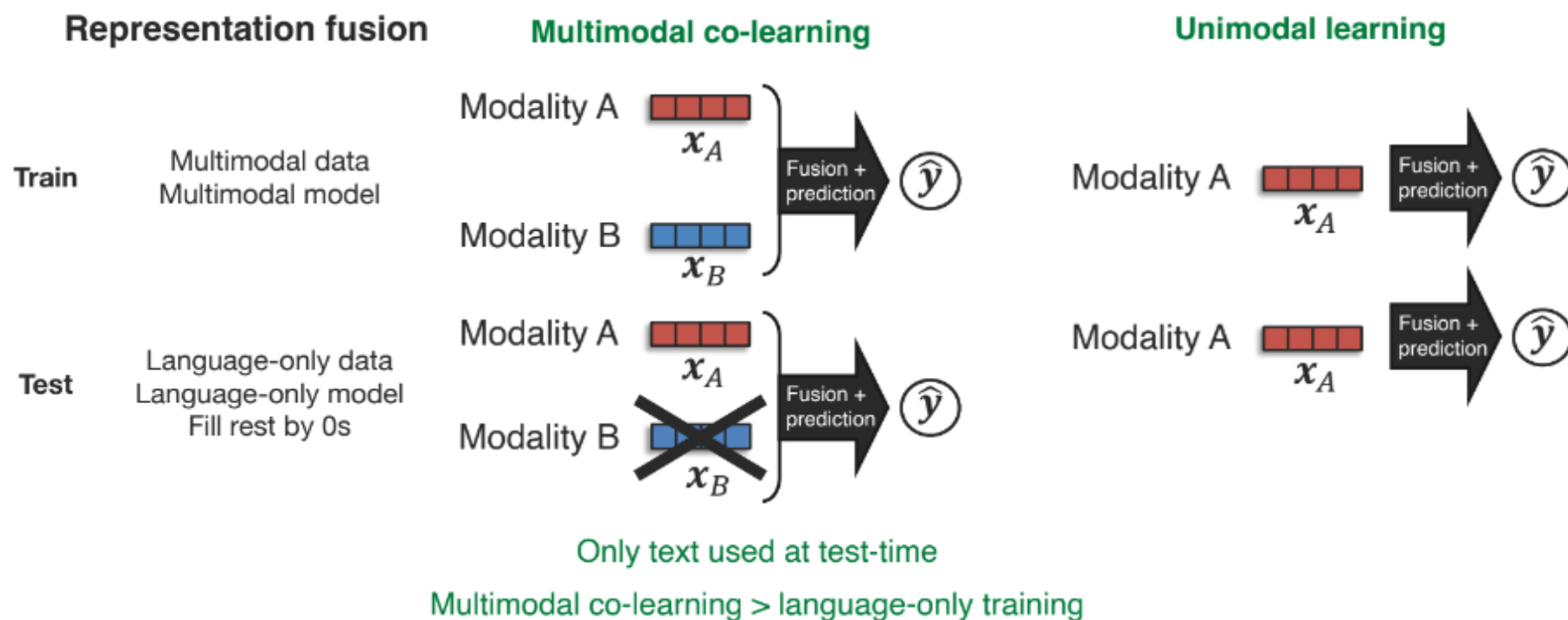
CO-LEARNING VIA REPRESENTATION

- Transferring information from secondary to primary modality by sharing representation spaces between both modalities.



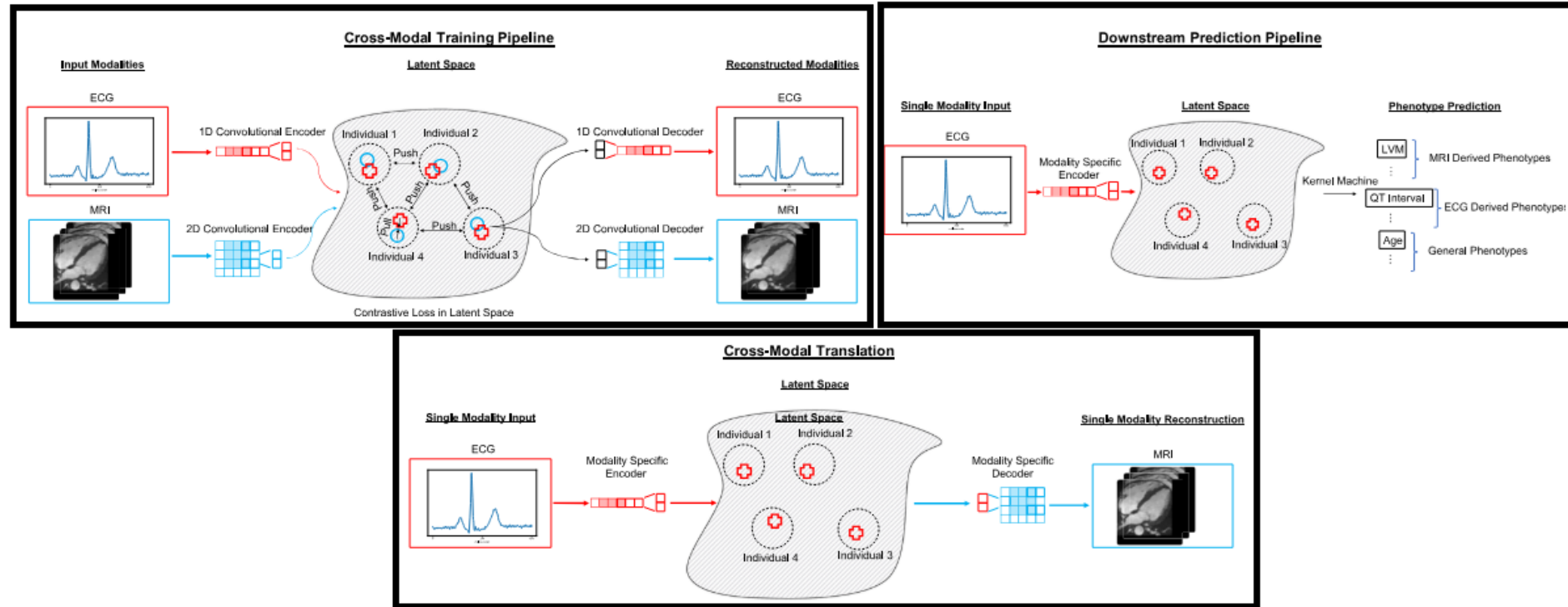
CO-LEARNING VIA REPRESENTATION

- Do not confuse multimodal co-learning with unimodal learning!!!



CO-LEARNING VIA REPRESENTATION

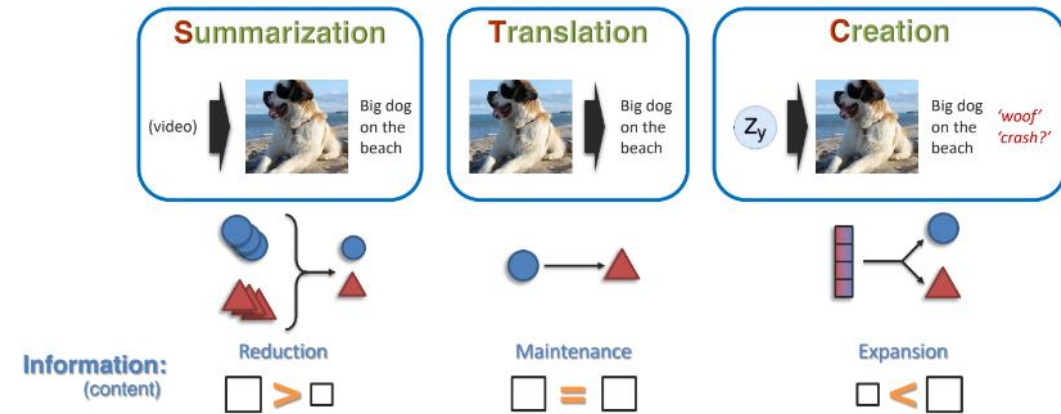
Ex: Cross-modal autoencoder framework for diagnosis



Radhakrishnan, Adityanarayanan, et al. "Cross-modal autoencoder framework learns holistic representations of cardiovascular state." *Nature Communications* 14.1 (2023): 2436.

GENERATION

- Generation was mentioned for the task of translation, but it also includes **summarization** and **creation**.
 - **Summarization:** Summarizing multimodal data to reduce information content while highlighting the most salient parts of the input. E.g., Image, video, audio guidance for text summarization
 - **Creation:** Simultaneously generating multiple modalities to increase information content while maintaining coherence within and across modalities.



SUMMARIZATION

- Wikipedia: the process of shortening a set of data computationally to create an abstract that represents the **most important** or **relevant information** within the original content.
- Formally: the process of obtaining the set $X_{sum} = f(D)$ such that $length(X_{sum}) \leq length(D)$, where X_{sum} is the output summary, D is the input data, and function $f(.)$ is the summarization function.

SUMMARIZATION

- **Multimodal summarization:** When the input dataset D can be broken down into several partially disjoint sets of different modality information $\{M_1, M_2 \dots M_n\}$, where $n \geq 2$ and $\exists$ several pairs of (M_i, M_j) for $(i, j) \in \{1, \dots, n\}$ such that the shared latent information between (M_i, M_j) is not $\emptyset$, then the task of obtaining the set $X_{sum} = f(D)$ is known as multi-modal summarization. If $n' \geq 2$ for $X_{sum} = \{M'_1 M'_2 \dots M'_{n'}\}$, then the output summary is multi-modal; otherwise, the output is a uni-modal summary.

SUMMARIZATION

- For any information processing task in multi-modal scenarios, including content summarization, amongst all the modalities, there is often a **preferable mode of representation** based on the significance and ability to fulfill the task.
- The majority of the works commonly utilize additional **complementary modalities** to enhance the feature representation for the **primary modality**, however, they typically generate summaries from a **single modality**.
 - Ex: building a language-guided framework to obtain a summary video conditioned on the text, learning joint representations of text and images, and outputting text summaries.



SUMMARIZATION

- Very recently multimodal summarization dataset with
 - text and image modalities [A],
 - video-based news articles (text + video) [B]
 - audio + transcript [C].

[A] Junnan Zhu, et al.: Multimodal summarization with multimodal output. In Proc. of the 2018 conference on empirical methods in natural language processing, pages 4154–4164, 2018.

[B] Mingzhe Li, et al. Vmsmo: Learning to generate multimodal summary for video-based news articles. arXiv preprint arXiv:2010.05406, 2020

[C] Xiyan Fu, et al., Mm-avs: A fullscale dataset for multi-modal summarization. In Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pages 5922–5926, 2021



GENERATION

- **Creation:** Simultaneously generating multiple modalities to increase information content while maintaining coherence within and across modalities.
 - Exp: language generation (image captioning), high-resolution speech and sound generation, and photorealistic image generation (text-to-image synthesis, speech-to-image generation)
 - Art, design, and storytelling, interactive museums
 - LLava, Mistral, and ImageBind **combine text and images**, allowing them to **generate captions** for pictures, answer questions about visual content, or even **create new images** based on a **textual** description.

CREATION

- Aims to generate novel high-dimensional data (which could span text, images, audio, video, and other modalities) from **small initial examples or latent conditional variables**.
- This conditional decoding process is extremely challenging since it needs to be
 - (1) conditional: preserve semantically meaningful mappings from the initial seed to a series of long-range parallel modalities,
 - (2) synchronized: semantically coherent across modalities,
 - (3) stochastic: captures many possible future generations given a particular state, and
 - (4) auto-regressive across possibly long ranges.



CREATION

- The biggest challenge facing multimodal generation is difficulty in **evaluating** generated content.
 - Serious ethical issues when fake news, hate speech, deepfakes, and lip-syncing videos can be easily generated.
- While the ideal way to evaluate generated content is through **user studies**, it is
 - time-consuming,
 - costly, and
 - can potentially introduce subjectivity bias into the evaluation process.

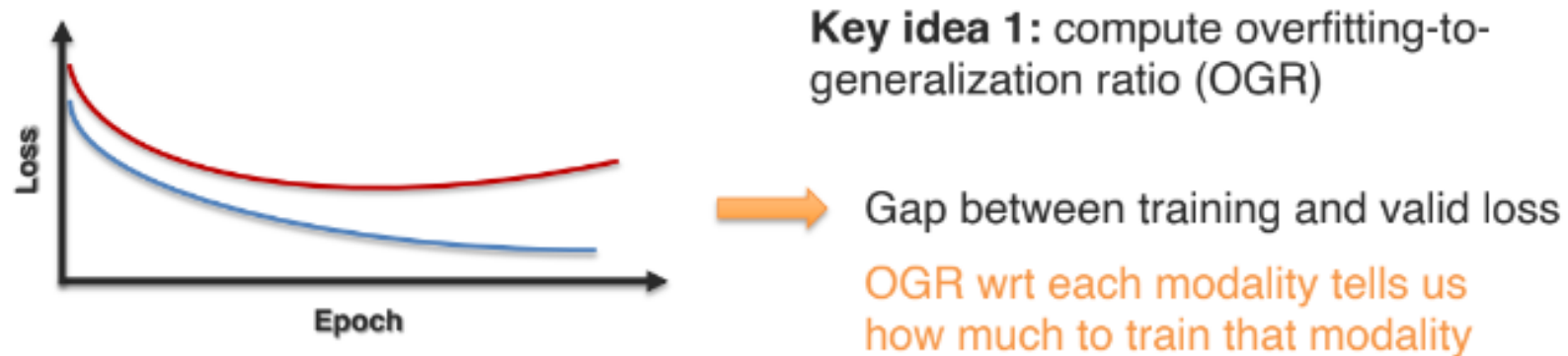


LEARNING AND OPTIMIZATION PROCESS

- Generalization across modalities and tasks
 - Many real-world platforms, such as cell phones, smart devices, self-driving cars, healthcare technologies, and robots, now integrate a much larger number of sensors beyond the prototypical text, video, and audio modalities.
 - Generalization across paired modality inputs and in unpaired scenarios where each task is defined over only a small subset of all modalities are still an ongoing challenges.

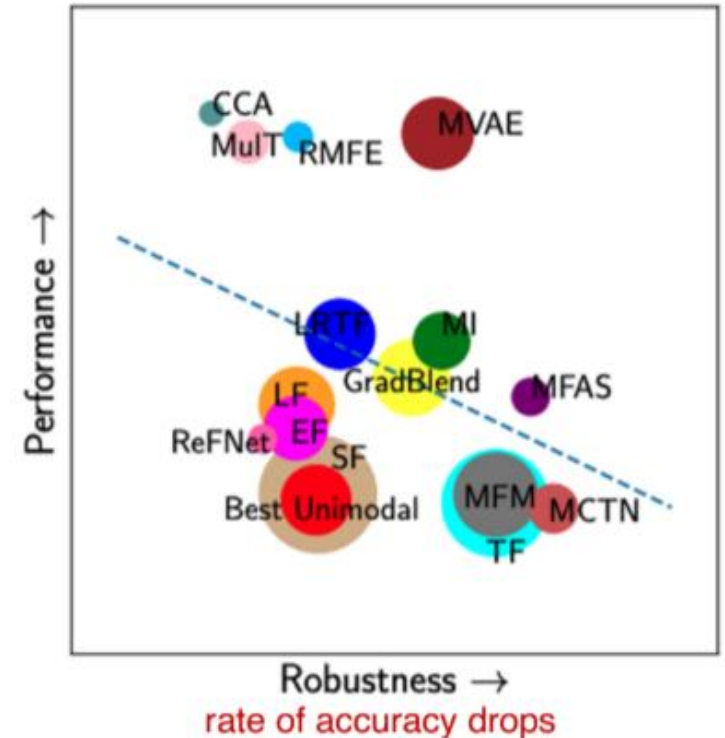
LEARNING AND OPTIMIZATION PROCESS

- Better optimization for balanced and efficient training
 - Multimodal networks are often prone to **overfitting** due to **increased capacity**, and different modalities **overfit** and **generalize** at **different rates** so training them jointly with a single optimization strategy is sub-optimal.
 - Some solutions by Wang et al. defining **OGR**



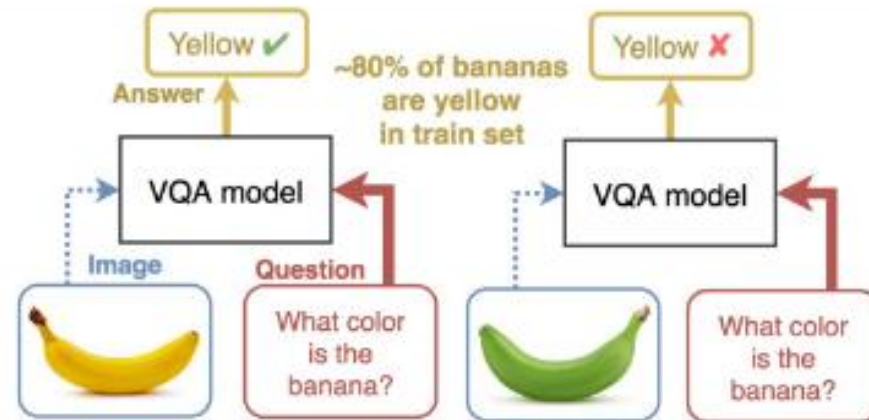
LEARNING AND OPTIMIZATION PROCESS

- Balancing the **tradeoffs** between **performance**, **robustness**, and **complexity** in real-world deployment.
 - **Performance**: The model's ability to make correct predictions or classifications.
 - **Robustness**: generalization, resilience to noise (the model's performance when faced with noisy, corrupted, or adversarial inputs), and fault tolerance (the model's capacity to maintain functionality despite unexpected inputs).
 - **Complexity**: Model size, inference time, scalability.



MODALITY BIASES

- Unimodal biases and modality collapse
 - Visual Question Answering models reply to the questions without looking at the image [A][B][C]



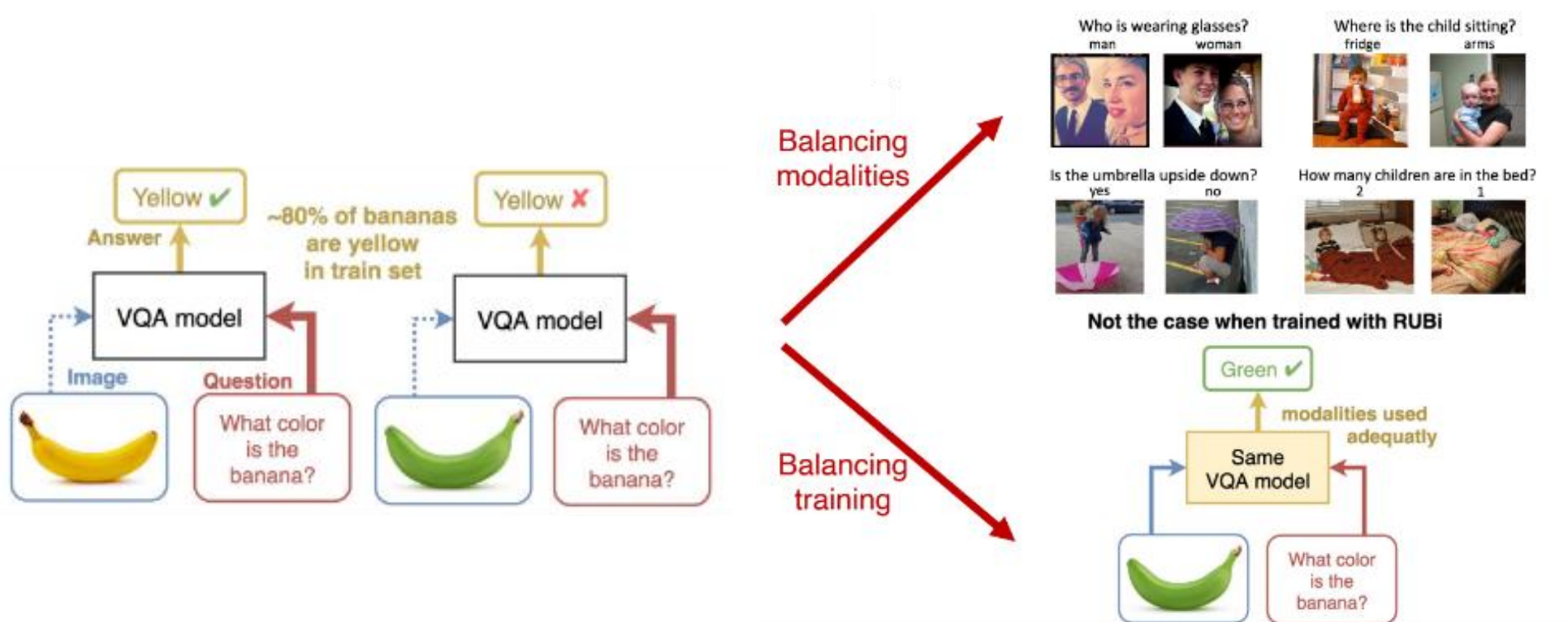
[A] Javaloy et al., Mitigating Modality Collapse in Multimodal VAEs via Impartial Optimization. ICML 2022]

[B] Wu et al., Characterizing and Overcoming the Greedy Nature of Learning in Multi-modal Deep Neural Networks. ICML 2022

[C] Goyal et al., Making the V in VQA Matter: Elevating the Role of Image Understanding in Visual Question Answering. CVPR 2017

MODALITY BIASES

- Unimodal biases and modality collapse
 - Possible solutions: a) Balancing modalities, b) Balancing training



Liang, Paul Pu, Amir Zadeh, and Louis-Philippe Morency. "Foundations & Trends in Multimodal Machine Learning: Principles, Challenges, and Open Questions." *ACM Computing Surveys* (2023).

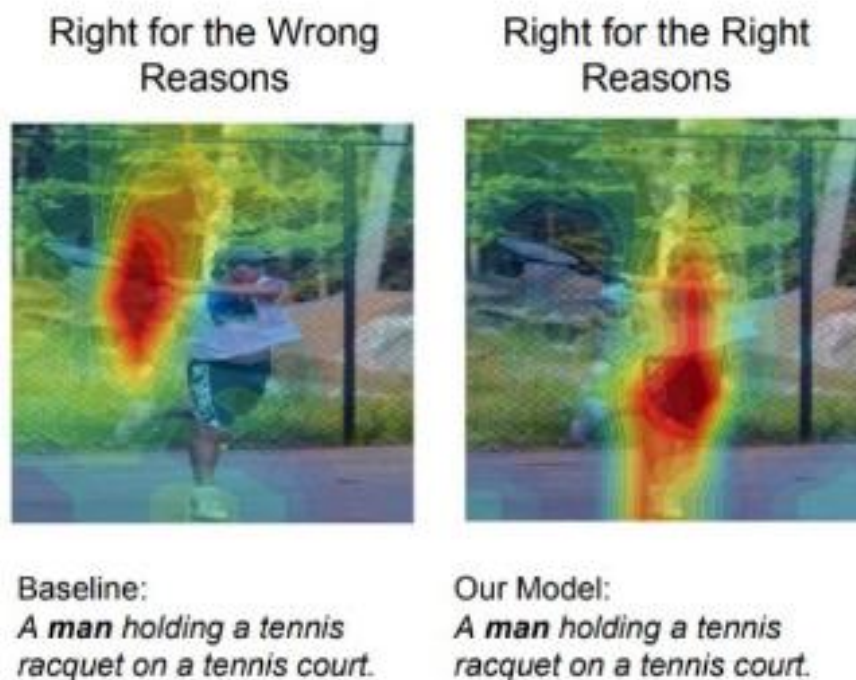
MODALITY BIASES

- **Fairness (and social biases)**
 - **Bias in Historical Data:** The data used to train a model can contain unfair or discriminatory patterns.
 - **Lack of Representation:** Insufficient data for certain groups can lead to inaccurate predictions and unfair treatment.
- Ex: Image captioning models capture spurious correlations between **gender**



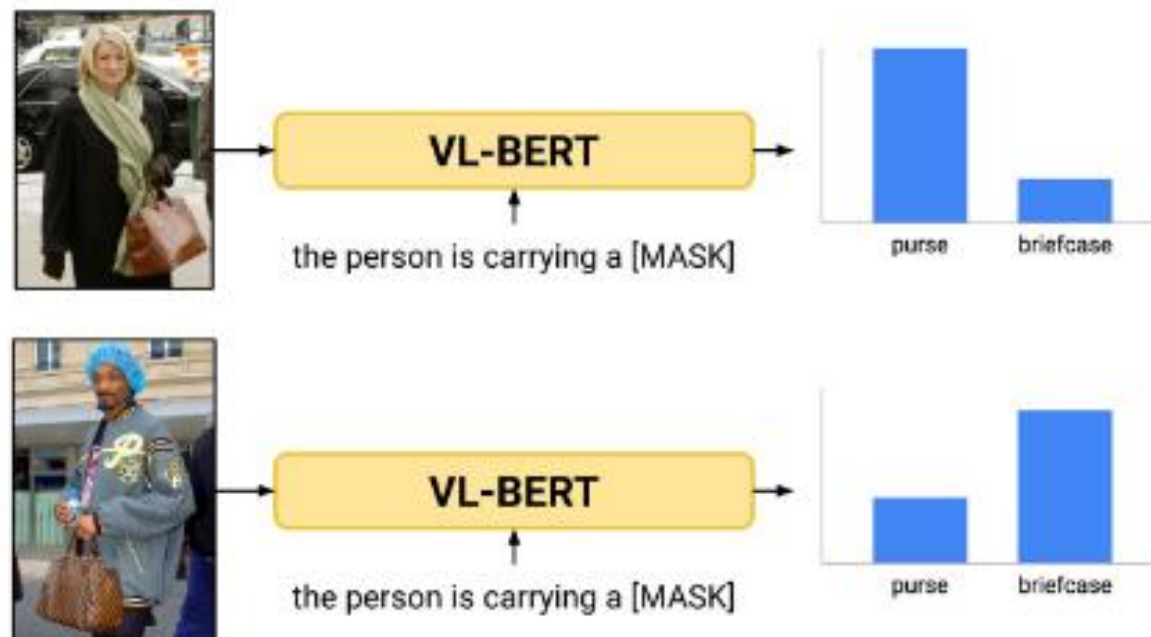
MODALITY BIASES

- Fairness and social biases
 - Ex: Image captioning models capture spurious correlations between generated actions



MODALITY BIASES

- Fairness and social biases: cross-modal interactions worsen social biases

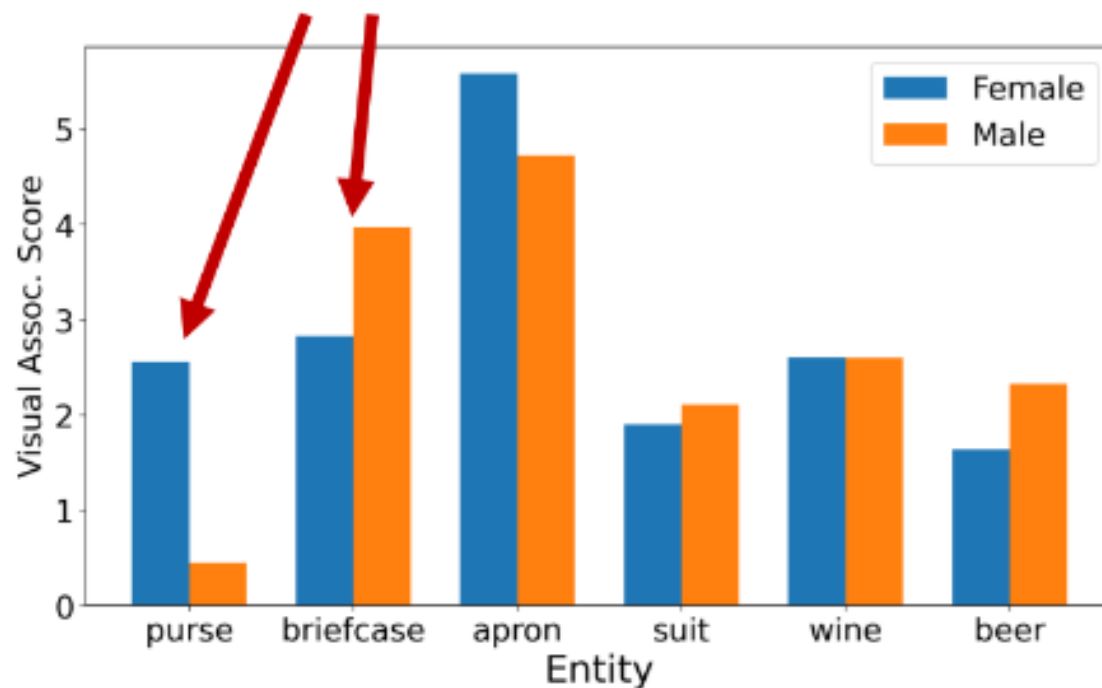


[Srinivasan and Bisk, Worst of Both Worlds: Biases Compound in Pre-trained Vision-and-Language Models. NAACL 2022]

MODALITY BIASES

- Fairness and social biases: cross-modal interactions worsen social biases

Visual information makes model more confident
in reinforcing gender stereotypes



[Srinivasan and Bisk, Worst of Both Worlds: Biases Compound in Pre-trained Vision-and-Language Models. NAACL 2022]



That's all

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